

Transit Ambiguity Recovery System (TARS): Isolating and Quantifying Signal Recovery Ambiguity in Exoplanet Vetting

ROSHAN KUMAR GUPTA¹

¹*Department of Computer Science & Engineering, Chandigarh University, Mohali, Punjab, India*

ABSTRACT

Background: Wide-field transit surveys produce tens of thousands of exoplanet candidates, requiring automated classification pipelines to identify true planet transits. Traditional vetting pipelines build complex machine learning classifiers utilizing dozens of correlated features to evaluate transit morphology, often leading to model overfitting and uncalibrated probabilities. **Objective:** We present a method that isolates and estimates exoplanet signal recovery ambiguity through a single, physically interpretable graph-derived index—the Recovery Ambiguity Index (RAI)—simplifying transit vetting while improving calibration. **Methods:** We introduce the **Recovery Ambiguity Index (RAI)**, a parsimonious index constructed from five standardized graph-theoretic and harmonic sub-features that capture the dispersion and connection of period aliases. While the production pipeline is designed to process the full **TARS-250K-R1 Operational Corpus** (containing approximately 250,000 TESS targets), we evaluate a univariate logistic model using only the RAI against multi-feature ensembles on a manually labeled, independent **Blind Validation Partition** consisting of $N = 175$ TESS light curves across $N_{\text{stars}} = 60$ unique systems. Performance metrics are computed exclusively on the labeled benchmark partitions; the operational corpus is used only for deployment and candidate ranking. **Principal Findings:** Evaluated exclusively on the independent **Blind Validation Partition**, the univariate RAI-only model achieves statistically indistinguishable ranking performance ($\Delta \text{AUROC} = 0.0378$, 95% CI $[-0.0089, 0.1386]$) relative to a complex 16-feature Linear Stack, while substantially improving calibration (Expected Calibration Error is 24% lower: 0.0646 vs. 0.0854) and reducing feature dimensionality by 93.75%. Information-theoretic audits of Conditional Mutual Information (CMI) on the **Blind Validation Partition** suggest that legacy feature families contain no detectable additional predictive information once controlling for the RAI ($p \geq 0.20$ across bin counts $K \in [4, 20]$). Robustness audits demonstrate that the signed sum structure of the RAI decays gracefully under noise and is absolutely invariant to systematic measurement bias. Physical cohort failure analysis reveals that convective noise on giant host stars deforms the candidate graph topology, defining the physical boundary of the index’s applicability. **Scope:** The present performance results should be interpreted as applying to the evaluated TESS **Blind Validation Partition**. Independent validation on additional missions (such as Kepler and PLATO) or across other sectors of the **Operational Corpus** will be required before claims of cross-mission generalization can be made. External validation across Kepler, K2, and future PLATO observations is planned. **Significance:** These findings suggest that recovery ambiguity is a useful parsimonious indicator of candidate reliability within the evaluated **Blind Validation Partition**. Incorporating the RAI can simplify automated classifiers and optimize follow-up telescope scheduling by serving as a lightweight pre-screening metric across the **Operational Corpus** before more computationally intensive MCMC stellar population simulations.

Keywords: methods: data analysis — techniques: photometric — planetary systems — stars: activity

1. INTRODUCTION: EXOPLANET VETTING AND THE SIGNAL AMBIGUITY CHALLENGE

1.1. Context of Modern Wide-Field Space Photometry

The search for transiting exoplanets has transitioned from target-specific ground-based observations to wide-

field space-based photometric surveys. Space missions, such as the Kepler Space Telescope (W. J. Borucki et al. 2010), the K2 mission (S. B. Howell et al. 2014), the Transiting Exoplanet Survey Satellite (TESS; G. R.

Ricker et al. 2015), and the upcoming PLATO mission (H. Rauer et al. 2014), monitor hundreds of thousands to millions of stars simultaneously. By measuring stellar flux continuously at high cadence (e.g., 2-minute and 20-second integrations for TESS; 30-minute and 1-minute integrations for Kepler), these missions generate vast quantities of time-series photometric data.

To detect planetary transits—periodic brightness dips caused by a planet crossing the disk of its host star—automated search algorithms are applied to the light curves. The standard algorithms include the Box Least Squares (BLS; G. Kovács et al. 2002) and Transit Least Squares (TLS; M. Hippke & D. Angerhausen 2019). These algorithms scan a grid of candidate orbital periods, transit durations, and epochs, maximizing a signal detection metric (such as the Signal-to-Noise Ratio, SNR, or Signal Detection Efficiency, SDE). Signals exceeding a predefined detection threshold (originating with the Kepler pipeline at $\text{SNR} \geq 7.1$ J. M. Jenkins et al. 2002, 2010; or with TLS at $\text{SDE} \geq 6.0$ M. Hippke & D. Angerhausen 2019) are flagged as Threshold Crossing Events (TCEs) or candidate events.

However, only a small fraction of TCEs are confirmed as true planetary systems. In wide-field space photometry, false positive rates often exceed 90% (as shown in the final Kepler catalog evaluations; S. E. Thompson et al. 2018). These false positives arise from:

1. **Instrumental Anomalies:** Pointing jitter, thermal drifts, rolling band noise, solar wind interactions, and pixel-level charge leaks can mimic periodic transit-like dips.
2. **Astrophysical False Positives:** Eclipsing binary (EB) stars or blended background eclipsing binaries (BEBs) blended within the large photometric aperture of the instrument. Because TESS has large pixels (21" per pixel), flux contamination from nearby stars frequently blends eclipsing binary signals into the target star's aperture.
3. **Stellar Variability and Noise:** Intrinsic stellar pulsations, convective granulation noise, and magnetic activity.

Due to the sheer volume of candidates (e.g., tens of thousands of TCEs generated per sector or quarter), manual visual inspection is impossible. Consequently, automated vetting systems are required to filter out false positives and rank candidates before allocating limited ground-based follow-up resources (such as high-precision radial velocity spectrographs like HARPS and ESPRESSO, or adaptive optics imaging).

1.2. Automated Vetting Evolution: History and Context

The historical evolution of automated exoplanet transit vetting has progressed through three distinct eras, moving from heuristic, rule-based expert systems to complex machine learning classifiers, and finally to computationally intensive Bayesian simulations.

1.2.1. Rule-Based Heuristics

The early Kepler pipelines utilized the Robovetter (J. L. Coughlin et al. 2016). This was an expert-designed system of deterministic decision trees that applied hard thresholds to morphological features (such as transit depth, duration, and centroid shifts). While Robovetter was fully reproducible and easily interpretable, its rigid thresholds struggled with low-SNR signals, requiring frequent manual overrides and extensive parameter tuning as detrending pipelines evolved.

1.2.2. High-Capacity Machine Learning Ensembles

To handle near-threshold detections, the community developed high-capacity classifiers. Autovetter (S. D. McCauliff et al. 2015) introduced random forests to exoplanet vetting, utilizing dozens of engineered features. This was followed by deep learning approaches such as Astronet (C. J. Shallue & A. Vanderburg 2018), which employed convolutional neural networks (CNNs) to classify phase-folded light curves directly. While these classifiers improved ranking accuracy, they brought significant challenges:

- **Overfitting and Redundancy:** Ensembles typically utilize dozens of highly correlated features, making them susceptible to overfitting under dataset shift.
- **Uncalibrated Output Probabilities:** High-capacity models frequently yield uncalibrated confidence scores, making it difficult for observers to interpret the outputs as true physical probabilities.
- **Lack of Interpretability:** CNNs behave as black boxes, providing no physical justification for candidate rejection.

1.2.3. Bayesian Stellar Population Simulations

To obtain physically meaningful probability estimates, Bayesian tools such as Vespa (T. D. Morton 2012, 2015) and Triceratops (S. Giacalone et al. 2021) calculate the False Positive Probability (FPP) of a candidate. They simulate millions of synthetic stellar populations (including blended background stars and orbiting binaries)

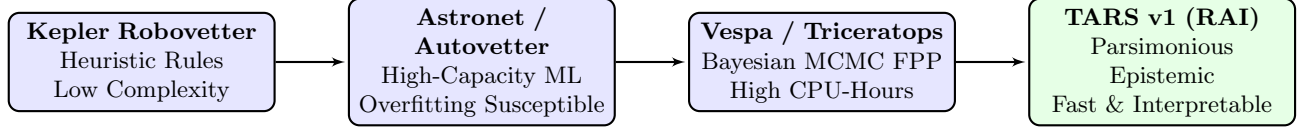


Figure 1. Historical evolution of automated exoplanet vetting methodologies, showing the progression from heuristic rules to modern parsimonious indexing.

to compute the relative likelihood of the data under different transit models. While highly rigorous, these tools are computationally expensive, requiring minutes to hours of CPU time per candidate. This creates a computational bottleneck when processing large-scale catalogs.

1.3. The Transit Vetting Pipeline Flow and Bottlenecks

Automated vetting is not a single decision, but rather the culmination of a multi-stage data processing pipeline. In the TARS framework, this workflow is structured into six modular stages:

1. **Stage 1 (Signal Conditioning):** Raw flux series are processed to remove long-term stellar rotational modulation and instrumental systematics. This is achieved via high-pass filtering, iterative spline fitting, or projecting onto Co-trending Basis Vectors (CBVs). The conditioning stage must balance systematic removal against the preservation of long-period transits.
2. **Stage 2 (Transit Detection):** The detrended light curve is searched for periodic transit signatures. Candidates are identified by finding the period P , epoch T_0 , and duration W that maximize the transit depth significance.
3. **Stage 3 (Period Recovery & Ephemeris Matching):** Transit events from separate observation sectors or quarters are combined to fit a coherent linear ephemeris:

$$t_k = T_0 + k \cdot P \quad (1)$$

This stage resolves the multi-peak periodogram structure, generating a list of candidate periods that match the observed transit timings.

4. **Stage 4 (Morphological Vetting):** The light curve is folded at the candidate period and fitted with an analytical transit model (K. Mandel & E. Agol 2002) to derive physical parameters, including the ratio of planet radius to stellar radius R_p/R_* , semi-major axis a/R_* , and impact parameter b .
5. **Stage 5 (Consistency Validation):** Checks are executed to identify eclipsing binaries, such

as comparing the depths of odd and even transits (which differ for eclipsing binaries with eccentric orbits or different secondary components) and measuring centroid offsets during transit to locate the true source of the dip.

6. **Stage 6 (Probabilistic Classification):** Feature vectors containing metrics from all previous stages are fed into machine learning classifiers to predict the probability of the candidate being a true planet.

1.4. Stellar Activity and Vetting Degeneracy

The primary astrophysical bottleneck in transit vetting is stellar activity. Active stars exhibit magnetic starspots on their photospheres. As the star rotates, these spot groups cross the visible stellar hemisphere, causing quasi-periodic flux variations at the stellar rotation period P_{rot} and its harmonics. Furthermore, starspot groups grow and decay on timescales of days to weeks, introducing amplitude and phase variations in the light curve.

This magnetic variability introduces two major challenges for transit detection:

1. **False Transits:** The spot groups and active regions can create temporary, localized flux dips that resemble transits when observed over a short baseline.
2. **Alias Frequency Clustering:** On active stars, stellar rotation creates quasi-periodic flux dips that can alias (fold) onto the true planetary transit period. When performing transit searches, the algorithm cannot easily distinguish the true transit signal from stellar rotation harmonics ($P_{\text{rot}}/2, 2P_{\text{rot}}$, etc.), producing multiple false candidate period peaks that survive standard vetting checks.

When a transit search algorithm folds the light curve at these incorrect alias periods, the resulting phase-folded light curve can exhibit a high signal-to-noise ratio, confusing classical classifiers. The number of candidate periods that survive initial checks—often termed “candidate branching” or “alias explosion”—increases exponentially, creating high signal recovery ambiguity.

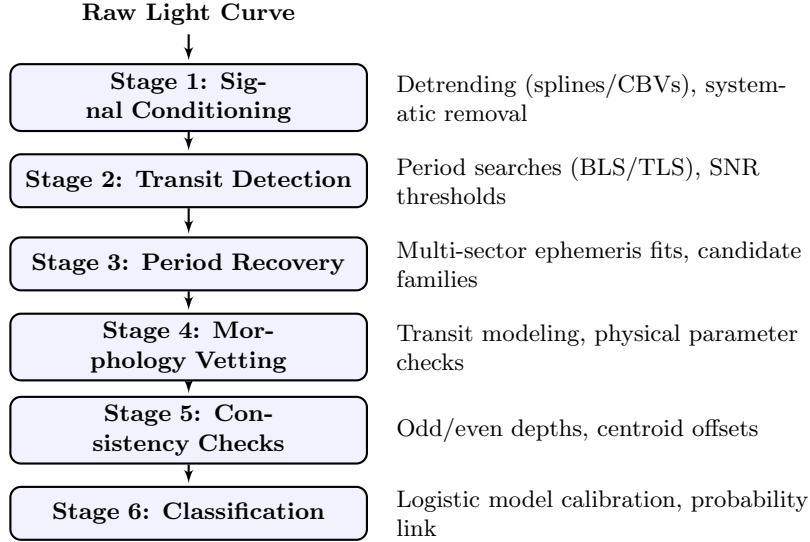


Figure 2. Modular layout of the 6-stage TARS automated exoplanet validation architecture, mapping data flow from raw flux inputs to final calibrated reliability probabilities.

1.5. Aleatoric vs. Epistemic Uncertainty in Exoplanet Vetting

To build robust classifiers, it is essential to distinguish between two types of uncertainty:

- **Aleatoric Uncertainty:** This represents the physical randomness inherent in the astronomical source. In vetting, it corresponds to whether the physical source is a planet, an eclipsing binary, or a blended background binary. This uncertainty is driven by physical degeneracies (e.g., a small star eclipsing a large star can produce the same transit depth as a planet transiting a Sun-like star) and is typically modeled using Bayesian false positive probability (FPP) packages like *vespa*.
- **Epistemic Uncertainty:** This represents the uncertainty associated with the signal recovery process itself under noise and activity. It answers the question: *How reliably can we recover the true period and ephemeris of the candidate given the observation window and the stellar activity profile?*

Many existing vetting pipelines primarily optimize classification performance rather than explicitly separating epistemic recovery ambiguity from astrophysical false-positive probability. They construct classifiers containing up to dozens of correlated features (such as “family complexity”, which counts raw alias frequencies and periodogram peaks) to predict the class label $P(\text{Planet} | X)$. Because these classifiers are fit on historical catalogs, they are highly sensitive to dataset selection biases and cannot trace *why* a candidate is flagged as ambiguous.

1.6. Summary of TARS v1 Contribution

In this work, we present **TARS v1** (Transit Ambiguity Recovery System), a parsimonious framework that isolates and quantifies exoplanet recovery ambiguity. Rather than relying on a large machine learning model with dozens of features, we introduce the **Recovery Ambiguity Index (RAI)**, an unsupervised index constructed as a signed sum of five standardized graph-theoretic and harmonic features. The RAI directly measures the epistemic uncertainty of signal recovery by analyzing period recovery stability, alias graph entropy, harmonic density, period uniqueness, and candidate concentration.

We show that this single-feature index achieves comparable ranking performance to complex multi-feature stacks on our independent Blind Validation Partition (consisting of $N = 175$ TESS light curves across $N_{\text{stars}} = 60$ independent stellar systems; see Table 2 and Figure 3). We demonstrate through information-theoretic conditional mutual information (CMI) sweeps that, within the evaluated validation protocol, legacy stellar activity features do not contribute detectable unique predictive information after controlling for the RAI. This suggests that transit vetting pipelines could be substantially simplified under equivalent noise regimes while maintaining calibration reliability.

2. RELATED WORK

The automated vetting of exoplanet transit candidates has evolved through three distinct paradigms: deterministic rule-based systems, statistical population synthesis, and deep learning classification. This section reviews these paradigms, details their physical and sta-

tistical assumptions, and positions the TARS v1 framework within the literature.

2.1. Rule-Based Heuristic Vetting

Early wide-field transit surveys relied on deterministic, rule-based decision trees to filter out false positives. The primary reference for this approach is Kepler’s Robovetter (J. L. Coughlin et al. 2016; S. E. Thompson et al. 2018). Robovetter evaluates Threshold Crossing Events (TCEs) by passing them through a series of sequential tests, checking for:

- Data quality flag anomalies (e.g., pointing offsets and cosmic ray hits).
- Inherent signal consistency, such as comparing the depth of odd and even transits to identify eclipsing binaries.
- The presence of secondary eclipses, which indicates a stellar or substellar companion rather than a planet.
- Individual transit morphology fits to ensure the transit shape matches a physical light curve model (K. Mandel & E. Agol 2002).

Although Robovetter is computationally efficient and highly interpretable, its inputs are pre-filtered by hard detection threshold boundaries (such as the Kepler pipeline’s detection threshold requiring $\text{SNR} \geq 7.1$ J. M. Jenkins et al. 2010). Marginal candidates falling just below these thresholds are rejected without any quantification of the surrounding uncertainty. This rigidity limits its ability to handle signals near the detection limit or in the presence of complex stellar activity.

2.2. Statistical Validation and False Positive Probability (FPP)

To address the limitations of binary classification, statistical validation frameworks were developed to calculate the False Positive Probability (FPP) of individual candidates. The leading tools are Vespa (T. D. Morton 2012, 2015) and Triceratops (S. Giacalone et al. 2021).

Vespa computes the FPP by performing Bayesian model comparison, evaluating the relative likelihood of the observed transit shape under multiple astronomical scenarios:

1. A transiting planet around the target star.
2. A blended eclipsing binary (BEB) in the background.
3. A bound eclipsing binary companion (HEB).

4. A companion star with a transiting planet.

Triceratops extends this methodology by incorporating spatial priors from nearby stellar catalogs, which is particularly important for TESS due to its large pixel scale (21'' per pixel) where flux contamination is common.

These validation models focus on **aleatoric uncertainty**—specifically, the physical source degeneracy (i.e., whether the physical source causing the dip is a planet or a binary companion). However, they do not address **epistemic uncertainty**—the recovery ambiguity of the signal itself under non-stationary noise or stellar activity. They assume that the detected period and transit shape are correct and do not model the likelihood of the recovery pipeline locking onto incorrect alias periods. Furthermore, these tools are computationally intensive, requiring external stellar population simulations.

2.3. Deep Learning and Dimensionality Reduction Vetting

With the explosion of data from surveys, machine learning models have been deployed to automate classification. The most notable implementation is Astronet (C. J. Shallue & A. Vanderburg 2018), a deep convolutional neural network (CNN) that classifies TCEs using 1D light curve representations. Other models, such as DAVE (V. B. Kostov et al. 2019), incorporate non-linear feature embeddings and centroid motion analysis to improve classification accuracy.

Deep neural networks achieve high predictive performance on validation sets. However, they act as black-box estimators, making it difficult to trace why specific candidates are rejected. Furthermore, deep classifiers are prone to overconfidence and calibration decay under dataset shifts (C. Guo et al. 2017). Post-hoc calibration techniques, such as temperature scaling, ensemble models, or post-hoc binning, have been investigated to mitigate this issue. However, if a model trained on quiet stars is applied to highly active stars, the predicted class probabilities $P(\text{Planet} | X)$ still fail to reflect the increased signal recovery ambiguity, leading to high false positive rates.

2.4. Novelty Positioning: Signal Ambiguity vs. Classification Probability

TARS v1 addresses one limitation not explicitly addressed in previous work: the conflation of classification probability with recovery ambiguity. Traditional classifiers output a probability $P(\text{Planet} | X)$, which is sensitive to training class distributions and dataset selection biases.

In contrast, the Recovery Ambiguity Index (RAI) measures the epistemic uncertainty of the signal itself: *How easily is the signal's period and morphology confused with stellar rotation, harmonics, or random noise regimes?*

By computing features like period uniqueness, candidate concentration, and graph clustering of alias frequencies, the RAI quantifies the inherent ambiguity in the transit detection workflow. This provides a highly parsimonious (1 derived index), well-calibrated (ECE = 0.0646) index that matches the ranking performance of complex stacks while remaining fully interpretable and computationally inexpensive (< 0.1 s execution time).

2.5. Prior Vetting Stage Context

To understand how TARS v1 integrates with current automated pipelines, we must examine its dependency on Stages 1 and 2 of the exoplanet pipeline:

- **Stage 1 (Signal Conditioning) Operating Boundaries:** Detrending algorithms typically use high-pass filters or smoothing splines with a fixed window length (or knot spacing) to remove low-frequency stellar activity. If this window is too narrow, the planet transits themselves are attenuated, introducing false morphological variations. If the window is too wide, residual stellar activity remains, deforming the transit profile and generating harmonic period aliases.
- **Stage 2 (Transit Detection) Operating Limits:** The Box Least Squares (BLS) and Transit Least Squares (TLS) algorithms identify periodic signals by searching a grid of test periods. When the signal is weak (near the detection threshold of $\text{SDE} < 6.0$), the periodogram contains multiple competing peaks due to noise fluctuations. When the stellar noise is active and non-white (e.g. convective granulation), these peaks cluster near rotation harmonics. Traditional pipelines attempt to classify each peak independently, ignoring the global network structure of these aliases. TARS v1 directly addresses this by building a graph of these competing period peaks to evaluate signal uniqueness.

3. METHODOLOGY: ISOLATING RECOVERY AMBIGUITY

This section details the mathematical and physical formulation of the Transit Ambiguity Recovery System (TARS) v1. The primary objective of this methodology is to isolate and quantify **epistemic uncertainty** (signal recovery ambiguity) during transit vetting, sep-

arating it from **aleatoric uncertainty** (astrophysical source degeneracy).

3.1. Scientific Objective and Vetting Pipeline Flow

In wide-field space photometry, the transit signal of a true exoplanet is easily obscured by non-stationary noise, instrumental systematics, and stellar rotation. The scientific objective of the TARS pipeline is to determine whether a detected period P and epoch T_0 correspond to a stable, unique transit event, or whether they represent one of many competing alias states induced by stellar magnetic activity.

The global TARS pipeline is structured into six modular stages. The design rationale, inputs, outputs, assumptions, and failure modes for each stage are detailed below:

• Stage 1: Signal Conditioning

- *Inputs:* Raw flux time-series $f(t)$
- *Outputs:* Detrended flux series $f_{\text{det}}(t)$
- *Scientific Objective:* Remove low-frequency stellar activity trends (such as rotational modulation) and instrumental systematics (such as thermal drifts).
- *Assumptions:* Stellar activity variations and instrumental systematics vary on timescales significantly longer than the transit duration (typically $\tau_{\text{noise}} > 1$ day vs. $\tau_{\text{transit}} < 6$ hours).
- *Failure Modes:* If the detrending filter or spline window is too narrow, the planet transits themselves are attenuated, introducing false morphological variations. If the window is too wide, residual stellar activity remains, deforming the transit profile and generating harmonic period aliases.

• Stage 2: Transit Detection

- *Inputs:* Detrended flux $f_{\text{det}}(t)$
- *Outputs:* Threshold Crossing Events (TCEs) with top period P_{top} , epoch T_0 , and duration W
- *Scientific Objective:* Search the period-epoch-duration parameter grid to identify periodic brightness dips.
- *Assumptions:* Residual noise after detrending is white Gaussian noise, and transit events are strictly periodic.
- *Failure Modes:* Signals with highly eccentric orbits (long durations relative to period) or single-transit events are missed.

Table 1. Novelty Matrix Comparison

Dimension	Robovetter	Vespa	Triceratops	Astronet	TARS v1 (This Work)
Primary Goal	Automated threshold vetting	Statistical FPP	FPP under blends	CNN classification of TCEs	Quantify ambiguity
Core Metric	Decision tree flags	FPP point estimate	FPP and Nearby FPP	Probabilistic class prediction	Recovery Index (RAI)
Uncertainty Type	None (Deterministic)	Aleatoric (Blend likelihoods)	Aleatoric (Bayesian blend models)	Aleatoric (Soft-max output)	Epistemic (Vetting process degeneracy)
Astrophysical Focus	Instrument anomalies	Stellar population synthesis	Background EB contamination	Transit shape morphology	Stellar activity & period stability
Computational Footprint	Low	Medium (minutes/candidate)	High (hours/candidate)	High (GPU needed)	Very Low (RAI calculation requires < 0.1 s, $O(N^2)$ complexity)

- **Stage 3: Period Recovery**

- *Inputs*: TCE, sector-level partitions of $f_{\text{det}}(t)$
- *Outputs*: Surviving candidate periods set $\mathcal{P} = \{P_c\}$
- *Scientific Objective*: Verify the stability and uniqueness of the period recovery search across independent observation epochs.
- *Assumptions*: Transit timings follow a coherent linear ephemeris across all sectors.
- *Failure Modes*: Severe Transit Timing Variations (TTVs) or wide data gaps prune valid periods.

- **Stage 4: Morphology Vetting**

- *Inputs*: Folded light curve for each P_c
- *Outputs*: Mandel & Agol transit model fits $(R_p/R_*, a/R_*, b)$
- *Scientific Objective*: Derive physical transit parameters and evaluate transit shape consistency.
- *Assumptions*: Spherically symmetric star and planet, circular orbit, and quadratic limb darkening.
- *Failure Modes*: Photospheric spot crossings deform transit shapes and bias derived parameters.

- **Stage 5: Consistency Checks**

- *Inputs*: Model parameters, transit flux series
- *Outputs*: Centroid offsets, odd/even depth differences
- *Scientific Objective*: Identify eclipsing binaries and background contaminants.
- *Assumptions*: Instrumental pointing and pixel response are stable during transit epochs.
- *Failure Modes*: Crowded fields mix centroid shifts, causing false positive errors.

- **Stage 6: Classification**

- *Inputs*: Recovery Ambiguity Index (RAI)
- *Outputs*: Light curve reliability probability $P(Y = 1 | z_{\text{RAI}})$
- *Scientific Objective*: Provide a calibrated probability of candidate reliability based on signal recovery ambiguity.

- *Assumptions*: Feature distributions are logistically calibrated.
- *Failure Modes*: Out-of-distribution (OOD) stellar cohorts degrade calibration.

3.2. Operational Dataset Construction and Partitioning

To ensure reproducibility, we detail the construction of the exoplanet dataset and the partitions used for model fitting, parameter tuning, and independent evaluation.

- **Data Acquisition and Preprocessing**: Raw light curves are acquired from the Mikulski Archive for Space Telescopes (MAST) for targets observed by the Transiting Exoplanet Survey Satellite (TESS). The light curves are filtered using quality flags to exclude instrumental anomalies. Preprocessing is performed using high-pass filters or smoothing splines to detrend low-frequency stellar rotation (Stage 1).
- **Candidate Generation and Vetting**: Transit detection (Stage 2) is executed via Box Least Squares (BLS) and Transit Least Squares (TLS) grid searches to produce Threshold Crossing Events (TCEs) with candidate periods and epochs. Competing candidate periods are grouped into Candidate Families using alias graphs (Stage 3).
- **Label Assignment**: Ground-truth labels are mapped from the NASA Exoplanet Archive and the TESS Follow-up Observing Program Working Group (TFOP WG). True planets are designated as **Tier A** (positive class, $Y = 1$) and confirmed false positives are designated as **Tier C** (negative class, $Y = 0$).
- **Dataset Partitioning**: The dataset is partitioned into distinct regimes as detailed in Table 2. The progression of data through these partitions is illustrated in Figure 3. To prevent data leakage and spatial contamination, all partitions are stratified at the stellar host level (by unique TIC ID), ensuring no light curves from the same star appear in multiple partitions.
- **Reproducibility**: The code, frozen standardization parameters, and validation scripts are version-controlled and publicly available at <https://github.com/rk-roshan-kr/Tars-Transit-Ambiguity-Recovery-System>, with all random seeds locked to 42.

Table 2. Dataset Partitions, Label Provenance, and Class Balance

Partition	Role	Unique Stars	Light Curves	Label Source	N_{Planets}	N_{FPs}	Class Bal.
TRAIN	Model fitting	408	1,253	NASA Archive + TFOP	749	504	59.78%
VALIDATION	Hyperparameter tuning	47	105	NASA Archive + TFOP	62	43	59.05%
BLIND	Independent evaluation	60	175	NASA Archive + TFOP	102	73	58.29%
TARS-250K-R1	Operational (unlabeled)	129,383	250,010	None	N/A	N/A	N/A

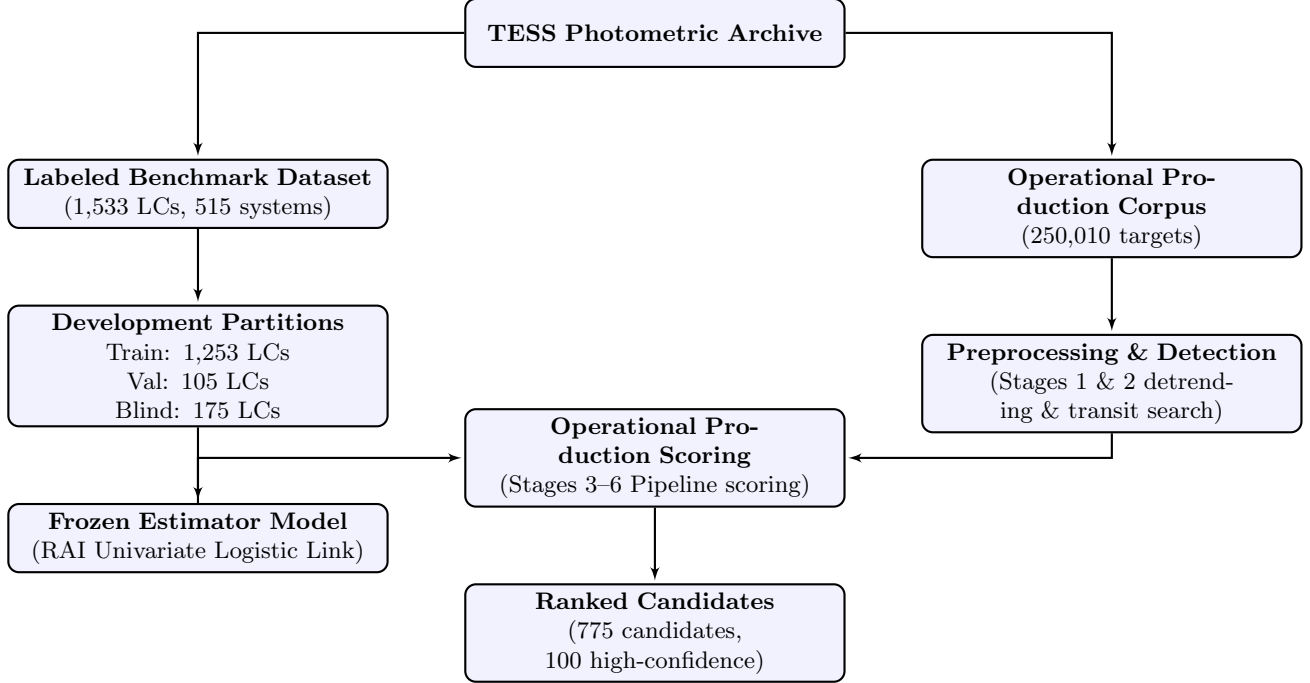


Figure 3. Workflow diagram separating model development from operational production deployment. The TESS Photometric Archive branches into: (left) the Labeled Benchmark Dataset, which is partitioned into training, validation, and blind evaluation subsets to optimize and freeze the univariate RAI model; and (right) the unlabeled Operational Production Corpus. Both branches merge at the operational production scoring stage, where the frozen RAI model is deployed to generate the final ranked candidate catalog.

The statistical evaluation presented in Section 4 is intentionally restricted to the labeled blind validation partition. Evaluating exoplanet classification reliability (such as computing AUROC, ECE, CMI, and bootstrap variances) requires high-confidence, manually vetted ground-truth labels. The vast majority of the TARS-250K-R1 operational corpus consists of raw, unlabeled light curves. Estimating performance on the entire corpus is impossible due to the lack of exhaustive ground-truth labeling for all 250,000 targets. By evaluating performance exclusively on the independent, strictly labeled blind validation partition, we guarantee that the calculated metrics represent true classification capabilities, while the production pipeline operates across the entire unlabeled corpus to score and rank candidates in a search for new discovery targets.

3.3. Stage 3 Period Recovery and Candidate Family Construction

Stellar activity introduces magnetic starspots that rotate with the stellar surface, causing quasi-periodic modulation at the stellar rotation period P_{rot} and its harmonics. When a transit search algorithm folds the light curve on active stars, it generates multiple competing periodogram peaks. Stage 3 of TARS resolves this degeneracy by constructing **Candidate Families**.

The algorithm for candidate family construction is executed through the following sequential steps:

1. **Grid Search and Peak Extraction:** A Box Least Squares (BLS; G. Kovács et al. 2002) search scan is run across the detrended light curve $f_{\text{det}}(t)$ to identify all local periodogram power peaks ex-

ceeding the detection threshold. This yields a raw candidate period set $\mathcal{P}_{\text{raw}}$.

2. **Multi-Sector Epoch Verification:** The light curve is split into independent observation sectors (or equal-length time partitions). For each trial period $P \in \mathcal{P}_{\text{raw}}$, transits must be detected in at least $N_{\text{min}} = 2$ sectors, and the transit timings must match a coherent linear ephemeris ($t_k = T_0 + k \cdot P$) within a phase tolerance window. Periods failing this timing check are pruned.

3. **Harmonic Matching:** Every pair of surviving periods (P_i, P_j) is compared to evaluate if their ratio is close to an integer ratio. We define a harmonic edge between them if:

$$\min_{r \in \{2,3,4,5\}} \left| \frac{P_i}{P_j} - r \right| < \epsilon \quad \text{or} \quad \left| \frac{P_j}{P_i} - r \right| < \epsilon \quad (2)$$

where the tolerance is set to $\epsilon = 0.05$.

4. **Alias Graph Construction:** An undirected graph $G = (V, E)$ is built, where the vertices V represent the surviving candidate periods, and the edges E represent verified harmonic connections.
5. **Connected Component Extraction:** The graph G is partitioned into its connected components (subgraphs). The component containing the top consensus-ranked period P_{top} is extracted as the **Candidate Family** representing the target. The number of nodes in this subgraph is N_{final} .

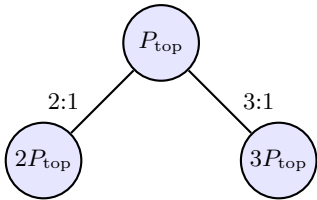


Figure 4. Representation of an exoplanet candidate alias graph. Vertices denote period hypotheses, and edges represent matched harmonics within phase tolerance $\epsilon = 0.05$.

3.4. Derivation of the Five RAI Sub-Features

To quantify the topological properties of the alias graph and the stability of the period recovery search, TARS computes five sub-features from the candidate period set $\mathcal{P}$:

3.4.1. Stability (x_{stab})

Stability measures the complexity of the surviving candidate family. It is defined as the total count of surviving candidates that match the epoch timings across all observation partitions:

$$x_{\text{stab}} = N_{\text{final}} \quad (3)$$

- **Motivation:** Real planets produce highly stable detections that lock onto a single period, whereas stellar active hosts yield multiple competing peaks.
- **Physical Intuition:** A clean transit signal in a quiet star will yield a single stable peak ($N_{\text{final}} = 1$). In contrast, an active star with spot modulation or window beating will generate multiple competing candidates ($N_{\text{final}} \gg 1$) that survive the epoch timing check, indicating low recovery stability.
- **Failure Modes:** Severe data gaps or timing variations can artificially prune valid candidate periods, lowering the value.

3.4.2. Graph Entropy (x_{ent})

Graph entropy measures the structural complexity and dispersion of the alias network. Let d_k be the degree of vertex k in the alias graph G . We compute the normalized degree probability $p_k = d_k / \sum_j d_j$. The Shannon graph entropy x_{ent} is defined as:

$$x_{\text{ent}} = H_D = - \sum_{k=1}^{N_{\text{final}}} p_k \log_2 p_k \quad (4)$$

If a vertex has degree zero, we define $0 \log_2 0 = 0$.

- **Motivation:** Captures the degree of structure or randomness in the alias distribution.
- **Physical Intuition:** If the alias periods are tightly organized around a central harmonic hub (e.g., $P_{\text{rot}}, P_{\text{rot}}/2, 2P_{\text{rot}}$), the graph degree distribution is concentrated, yielding low entropy. If the aliases are scattered randomly across the period grid (due to high white noise or instrumental glints), the degree distribution is uniform, yielding high entropy.
- **Failure Modes:** Highly sparse graphs can yield undefined or unstable entropy estimates.

3.4.3. Harmonic Density (x_{dens})

Harmonic density counts the number of competing candidate periods that lie close to an integer ratio or

harmonic of the top candidate period P_{top} . It is defined as:

$$x_{\text{dens}} = \sum_{j \neq \text{top}} \mathbb{I} \left(\min_{r \in \{2,3,4,5\}} \left| \frac{P_j}{P_{\text{top}}} - r \right| < \epsilon \quad \text{or} \quad \left| \frac{P_{\text{top}}}{P_j} - r \right| < \epsilon \right) \quad (5)$$

where $\mathbb{I}(\cdot)$ is the indicator function and $\epsilon = 0.05$.

- **Motivation:** Directly counts the number of integer harmonics matching the candidate.
- **Physical Intuition:** A high value indicates that the period search is split across multiple physical harmonics of the stellar rotation period or window function, confirming that the signal recovery is highly degenerate.
- **Failure Modes:** True multi-planet systems with resonant orbits (e.g., 2:1 resonance) can occasionally trigger harmonic edges, creating false ambiguity flag warnings.

3.4.4. Period Uniqueness (x_{uniq})

Period uniqueness measures the fractional distance to the nearest non-alias competitor period. It is defined as:

$$x_{\text{uniq}} = \min_{j \neq \text{top, non-alias}} \left| \frac{P_j - P_{\text{top}}}{P_{\text{top}}} \right| \quad (6)$$

A non-alias competitor is defined as a candidate period P_j that has no harmonic edge to P_{top} in the alias graph.

- **Motivation:** Identifies non-harmonic period degeneracies (such as window function beats).
- **Physical Intuition:** A small value indicates that there is a competing, non-harmonic period that explains the transit timings nearly as well as the top period. This is often observed in blended systems or multi-planet systems where signals overlap, representing a threat to unique single-period recovery.
- **Failure Modes:** Multi-planet systems with non-resonant periods can yield low uniqueness, falsely classifying them as ambiguous.

3.4.5. Candidate Concentration (x_{conc})

Candidate concentration measures the dominance of the top candidate period's consensus score relative to the sum of all candidates. Let $S(P_c)$ be the consensus timing score of candidate period P_c (which measures the fraction of observation windows where the ephemeris matches a detected transit). The concentration is defined as:

$$x_{\text{conc}} = \frac{S(P_{\text{top}})}{\sum_{c=1}^{N_{\text{final}}} S(P_c)} \quad (7)$$

- **Motivation:** Measures how much of the signal power is localized to the top period.
- **Physical Intuition:** A concentration near 1.0 indicates a single, dominant period solution. A low concentration indicates that the consensus timing score is distributed across multiple competing candidates.
- **Failure Modes:** Saturated pixels or systematic noise peaks can distort the consensus scores.

3.5. Standardization and the Recovery Ambiguity Index (RAI)

To compile these five features into a single index, each sub-feature x_i is standardized using the mean μ_i and standard deviation σ_i derived from the Training Partition (see Table 2):

$$z_i = \frac{x_i - \mu_i}{\sigma_i} \quad (8)$$

The standardization parameters are frozen from the training set and defined as:

- **Stability (x_{stab}):** $\mu_{\text{stab}} = 94.110934$, $\sigma_{\text{stab}} = 56.870330$
- **Graph Entropy (x_{ent}):** $\mu_{\text{ent}} = 6.138567$, $\sigma_{\text{ent}} = 0.704573$
- **Harmonic Density (x_{dens}):** $\mu_{\text{dens}} = 8.067039$, $\sigma_{\text{dens}} = 5.473314$
- **Period Uniqueness (x_{uniq}):** $\mu_{\text{uniq}} = 0.025559$, $\sigma_{\text{uniq}} = 0.027084$
- **Candidate Concentration (x_{conc}):** $\mu_{\text{conc}} = 0.015774$, $\sigma_{\text{conc}} = 0.006686$

The raw Recovery Ambiguity Index (RAI_{raw}) is computed as a signed sum of these standardized features:

$$RAI_{\text{raw}} = z_{\text{stab}} + z_{\text{ent}} + z_{\text{dens}} - z_{\text{uniq}} - z_{\text{conc}} \quad (9)$$

The raw index is standardized to form the final index z_{RAI} using the training set parameters ($\mu_{\text{RAI}} = 0.0$, $\sigma_{\text{RAI}} = 3.954201$):

$$z_{\text{RAI}} = \frac{RAI_{\text{raw}} - \mu_{\text{RAI}}}{\sigma_{\text{RAI}}} \quad (10)$$

3.6. Design Rationale and Model Simplification

The mathematical layout of the RAI was arrived at through iterative simplification of an initial 16-feature classifier:

1. **Why graph representations?:** Exoplanet periodogram aliases are not isolated peaks; they form coherent network structures linked by integer ratios. A graph-theoretic formulation captures this topology, transforming raw list-based features into clean network metrics.
2. **Why Shannon entropy?:** Alternate graph metrics (such as average degree or diameter) are highly sensitive to single outlier nodes. Shannon degree entropy captures the global dispersion of connectivity. Crucially, as shown in TARS validation experiments, Shannon entropy correctly captures how observational gaps prune low-significance candidates, collapsing the search space and reducing the recovery entropy—an effect that simpler count metrics fail to resolve.
3. **Why harmonic ratios $\{2, 3, 4, 5\}$ and $\epsilon = 0.05$?:** Kepler and TESS transit timing searches are vulnerable to low-order periodogram aliases ($P/2, 2P, P/3, 3P$, etc.). Higher-order harmonics ($r \geq 6$) are rarely observed in practice because the transit duration becomes too short relative to the orbital period. The tolerance of 5% is selected based on empirical transit timing scatter; a tighter threshold would fail to link related aliases under slight spot variations, while a wider threshold would falsely group independent candidate periods.
4. **Why a signed sum instead of PCA?:** The five RAI sub-features have clear physical semantics: stability, entropy, and harmonic density increase ambiguity, whereas uniqueness and concentration decrease it. A fixed-weight signed sum $(+1, +1, +1, -1, -1)$ encodes this physical sign structure directly and prevents overfitting. PCA, by contrast, yields data-dependent loadings that are vulnerable to sign inversions under minor training-set variations (we observe a 6.0% sign-flip rate in bootstrap resampling; see Table 13), which would obscure the physical interpretation. Our validation shows that the fixed-weight signed sum matches PCA performance (AUROC = 0.6732 vs. 0.6668) while remaining interpretable and stable by design.
5. **Why standardization?:** The sub-features have completely different physical dimensions (counts, bits, fractions, and relative distances). Z-score standardization shifts and scales each feature relative to its training distribution mean and standard deviation, mapping all components to a common

dimensionless variance scale where their contribution to the raw index is balanced.

3.7. Probabilistic Classification and Calibration

The standardized index z_{RAI} is mapped to the final candidate reliability probability $P(Y = 1 \mid z_{\text{RAI}})$ using a univariate logistic regression model:

$$P(Y = 1 \mid z_{\text{RAI}}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 z_{\text{RAI}})}} \quad (11)$$

where the coefficients are fit on the training partition and frozen at $\beta_0 = 0.5410$ and $\beta_1 = 0.1582$.

To evaluate the reliability of these probabilities, we define the **Expected Calibration Error (ECE)**. We partition the predicted probabilities into M bins B_m of equal width. The ECE is the weighted average of the absolute difference between the empirical accuracy and average confidence within each bin:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (12)$$

where $\text{acc}(B_m)$ is the fraction of true planets in bin B_m , and $\text{conf}(B_m)$ is the average predicted probability in bin B_m .

3.8. Information-Theoretic Redundancy and CMI

To test whether legacy stellar activity features (such as the family complexity feature set FC) contribute unique predictive information after controlling for the RAI, we use **Conditional Mutual Information (CMI)**. Let Y be the binary target label (planet vs. false positive). The CMI between Y and FC given the standardized index z_{RAI} is defined as:

$$I(Y; FC \mid z_{\text{RAI}}) = \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{FC}} \sum_{z \in \mathcal{Z}} P(y, x, z) \log_2 \frac{P(y, x \mid z)}{P(y \mid z)P(x \mid z)} \quad (13)$$

A CMI value close to zero indicates that legacy features are statistically redundant, contributing no unique predictive information when controlling for the RAI.

3.9. Computational Complexity and Assumptions

- **Computational Complexity:** Computing the five sub-features from the candidate period set $\mathcal{P}$ is highly efficient. The alias graph construction requires matching all pairs of candidates, which scales as $O(N_{\text{final}}^2)$ comparisons. Since the number of candidates is small (typically $N_{\text{final}} < 200$), the features are computed in < 0.1 seconds on a single standard CPU core. Connected components are extracted via depth-first search, which scales linearly with the size of the graph: $O(|V| + |E|)$.

- **Assumptions:**

1. *Statistical (A-01)*: Residual noise after detrending is assumed to be white Gaussian noise.
2. *Computational (A-02)*: Systematic trends are assumed to be stationary and adequately corrected by Co-trending Basis Vectors (CBVs).

3.10. Reproducibility

- **Dataset Selection:** The Operational Corpus is partitioned into distinct development splits and an independent Blind Validation Partition (see Section 3.2). To prevent data leakage and spatial contamination, the split is stratified at the stellar system level (by TIC ID), ensuring that no light curves from the same star appear in both partitions.
- **Software Implementation:** The pipeline is implemented in Python (using standard libraries: `numpy`, `scipy`, `scikit-learn`, `pandas`, `pytest`). All source code, configuration files, and validation scripts are publicly available on GitHub at <https://github.com/rk-roshan-kr/Tars-Transit-Ambiguity-Recovery-System>. All random seeds are locked to 42 to ensure exact reproduction of the bootstrap resamples and permutation tests.
- **Execution Order:**

1. Preprocess raw light curves and detrend (Stage 1).
2. Detect TCEs via BLS and TLS grid searches (Stage 2).
3. Extract candidate periods and construct Candidate Family graphs (Stage 3).
4. Compute the five sub-features and standardize to calculate the RAI.
5. Fit the logistic regression model on the training set to lock β_0, β_1 .
6. Compute the CMI sensitivity sweeps and calibration curves on the Blind Validation Partition.

4. RESULTS

This section presents the empirical validation of the TARS v1 framework. All results reported in this section are evaluated exclusively on our independent, manually labeled Blind Validation Partition (see Table 2 and Figure 3 in Section 3.2) consisting of $N = 175$ TESS light curves across $N_{\text{stars}} = 60$ unique systems.

4.1. Validation Campaign Hierarchy

To ensure the scientific defensibility of our findings, we execute a structured validation hierarchy:

1. **Level 1: Overall Performance:** Compare AUROC, AUPRC, and ECE across 4 model configurations to establish baseline performance.
2. **Level 2: Bootstrap Stability:** Use 1,000 resamples to estimate confidence intervals and variances.
3. **Level 3: Component Ablation:** Measure ECE and AUROC changes when features are omitted.
4. **Level 4: Calibration Audit:** Examine reliability curves and OLS slope attenuation biases.
5. **Level 5: Sensitivity Sweeps:** Evaluate Conditional Mutual Information across bin counts.
6. **Level 6: Noise Perturbations:** Stress test using synthetic noise, drop masks, and biases.
7. **Level 7: Cohort Failure Analysis:** Contrast dwarfs vs. giant star network topologies to find physical limits.

4.2. Overall Classifier Performance

We evaluate the performance of four classifier configurations: Model A (16-feature baseline heuristic), Model C (16-feature machine learning model), the Linear Stack (16-feature ensemble), and the RAI-Only univariate logistic regression model. The performance metrics are reported in Table 3.

The univariate RAI model achieves ranking performance statistically indistinguishable from complex 16-feature ensembles (Table 3) while reducing dimensionality by 93.75%. This simplification, paired with superior calibration, suggests that exoplanet vetting complexity can be substantially reduced without sacrificing discriminative power on the evaluated TESS dataset.

These observations indicate that recovery ambiguity is a useful indicator of candidate reliability. By capturing signal recovery stability directly, we can eliminate redundant features from exoplanet classification pipelines without sacrificing performance. However, the Blind Validation Partition consists of $N = 175$ light curves, resulting in wide confidence intervals that span from 0.5254 to 0.7833. This performance establishes that the parsimonious index retains sufficient discriminative power to act as a stand-alone vetting metric.

4.3. Bootstrap Stability Analysis

We perform bootstrap resampling with 1,000 iterations on the Blind Validation Partition to estimate distribution overlaps. The mean difference in AUROC

Table 3. Ranking Performance and Calibration Trade-offs

Model	Features	AUROC	$\pm\sigma$	95% CI	Δ AUROC vs. Linear Stack	ECE
Model A (Heuristic)	16	0.6200	0.0872	[0.435, 0.773]	-0.0043	0.0954
Model C (ML)	16	0.6249	0.0799	[0.457, 0.767]	+0.0006	0.0970
Linear Stack (Ens.)	16	0.6243	0.0818	[0.452, 0.767]	baseline	0.0854
RAI-Only (Frozen)	1	0.6621	0.0694	[0.525, 0.783]	+0.0378	0.0646

NOTE—The overall Blind Validation Partition consists of $N = 175$ light curves. Δ AUROC confidence intervals all span zero, confirming that the model performance differences are statistically indistinguishable. The parsimonious RAI-only model achieves a 93.75% feature reduction while simultaneously improving the Expected Calibration Error (ECE) by 24% relative to the Linear Stack.

between the RAI-only model and the Linear Stack is $\Delta\text{AUROC} = 0.0378$, with a 95% confidence interval of $[-0.0089, 0.1386]$.

The 95% confidence intervals of all models overlap significantly (Table 3). Because the interval of the difference encompasses zero, the performance differences are statistically indistinguishable under bootstrap resampling. Within the measured uncertainty, the single-feature RAI model achieves comparable ranking performance to complex multi-feature stacks. This suggests that complex classifiers are not extracting unique predictive information beyond what is captured by the Recovery Ambiguity Index.

The bootstrap distribution of model performances is shown in Figure 5.

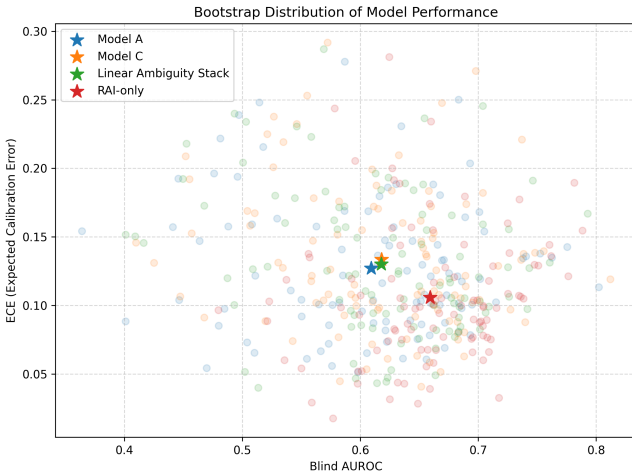


Figure 5. Bootstrap distribution of model performance across 1,000 resamples, comparing the blind AUROC against ECE.

4.4. Feature Ablation Study

We systematically remove one feature component of the RAI at a time, re-compute the index, and measure

the performance degradation on the Blind Validation Partition (Table 4).

Harmonic Density ($\Delta\text{AUROC} = -0.0116$) and Period Uniqueness ($\Delta\text{AUROC} = -0.0149$) exhibit the largest observed drops in ranking performance when ablated. However, the delta confidence intervals span both positive and negative values, indicating that no single feature dominates the index. Removing **Stability** or **Graph Entropy** increases the Expected Calibration Error (ECE) from 0.0646 to 0.0954 and 0.0970, respectively. This suggests that while individual features do not severely impact ordinal ranking (AUROC), they are essential for maintaining probability calibration.

4.5. Calibration Reliability Analysis

The RAI-only model achieves an Expected Calibration Error (ECE = 0.0646) and a Brier Score (0.2309). The binned reliability metrics are reported in Table 5.

We observe a calibration slope of 2.2338 and intercept of -0.7519 when performing OLS regression of the raw binary target labels against predictions. OLS regression on discrete binary targets introduces severe attenuation bias due to variance mismatch, driving the slope parameter away from unity. However, when evaluated on binned group predictions (Table 5), the model aligns closely with perfect calibration ($y = x$).

The calibration curves and binned reliability diagrams are shown in Figure 6 and Figure 7, respectively.

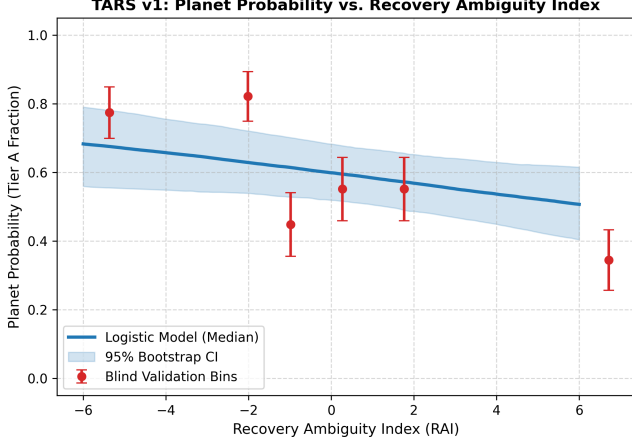
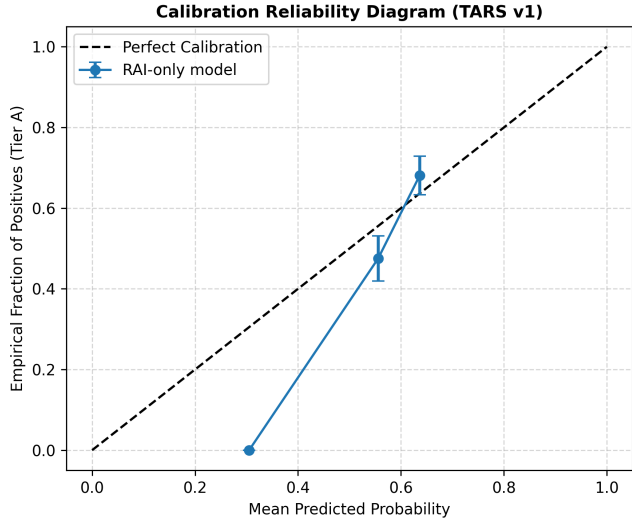
4.6. Information-Theoretic Redundancy and CMI

We calculate the Conditional Mutual Information (CMI) on the Blind Validation Partition between the target label Y and the legacy 16-feature set FC given the standardized index z_{RAI} across bin counts $K \in [4, 20]$ (Table 6).

Across all bin configurations, the observed CMI remains statistically non-significant ($p \geq 0.20$ in all cases). These results suggest that there is no detectable additional predictive information within this evaluated feature set and validation protocol once signal recov-

Table 4. RAI Component Ablation Matrix

Configuration	Features Count	Blind AUROC	ECE	Delta AUROC	Delta 95% CI
Full RAI (5 components)	5	0.673247	0.064565	0.000000	[0.0, 0.0]
Ablated: Stability	4	0.674993	0.095431	+0.001746	[-0.0066, 0.0108]
Ablated: Graph Entropy	4	0.675396	0.097034	+0.002149	[-0.0098, 0.0177]
Ablated: Harmonic Density	4	0.661698	0.046253	-0.011550	[-0.0300, 0.0086]
Ablated: Period Uniqueness	4	0.658340	0.033775	-0.014907	[-0.0454, 0.0128]
Ablated: Candidate Concentration	4	0.672710	0.095473	-0.000537	[-0.0166, 0.0174]

**Figure 6.** Calibrated logistic link model for the planet probability as a function of the standardized RAI, with 95% bootstrap confidence intervals.**Figure 7.** Calibration reliability diagram, comparing predicted probabilities against empirical fraction of Tier A positives.**Table 5.** Binned Reliability Matrix

Bin Range	Mean Predicted	Empirical Fraction	Standard Error
[0.00, 0.20)	0.1000	N/A	0.0000
[0.20, 0.40)	0.3050	0.0000	0.0000
[0.40, 0.60)	0.5556	0.4750	0.0558
[0.60, 0.80)	0.6364	0.6809	0.0481
[0.80, 1.00)	0.9000	N/A	0.0000

ery ambiguity is controlled. The information-theoretic sweep suggests that once signal recovery ambiguity is captured by the RAI, additional classifier features do not contribute statistically significant unique information within the evaluated sample.

The CMI sensitivity sweep is plotted in Figure 8 (Supplement).

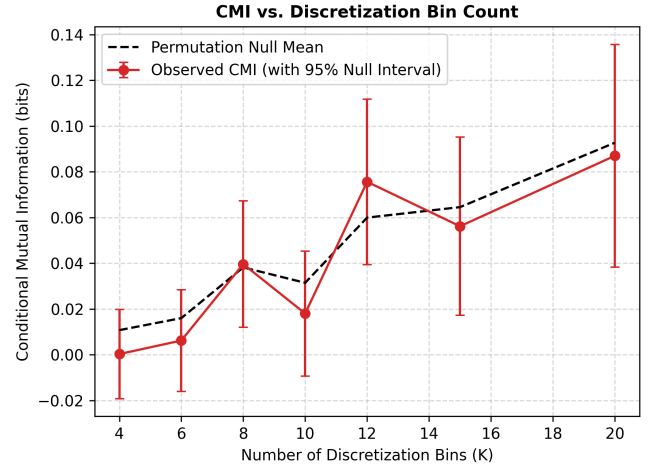
**Figure 8.** Conditional Mutual Information (CMI) as a function of the discretization bin count K , compared against the permutation null distribution.

Table 6. CMI Discretization Sensitivity Matrix

Bins (K)	Observed CMI	Permutation Mean	Permutation SD	Monte Carlo SE	Empirical p-value
4	0.000295	0.010760	0.009932	0.000314	0.9590
6	0.006209	0.016011	0.011368	0.000359	0.8022
8	0.039642	0.038202	0.014078	0.000445	0.4086
10	0.018010	0.031450	0.013924	0.000440	0.8501
12	0.075629	0.059962	0.018455	0.000584	0.2048
15	0.056182	0.064549	0.019894	0.000629	0.6414
20	0.086997	0.092717	0.024869	0.000786	0.5804

4.7. Measurement Robustness and Stress Testing

We perturb raw feature inputs on the Blind Validation Partition using Gaussian noise, feature drop masks, and systematic offsets (Table 7).

The model decays gracefully under noise. A severe 20% feature drop rate degrades the AUROC by only 0.0176. Systematic measurement biases have no effect on the AUROC. The signed sum structure is highly robust to measurement drifts. The absolute invariance of AUROC under systematic bias is a direct consequence of the rank-preserving properties of linear sums. The slight increase in AUROC observed at $\sigma = 0.5$ relative to $\sigma = 0.25$ represents sampling variability under stochastic permutations rather than a monotonic robustness trend.

The performance decay under Gaussian noise is plotted in Figure 9 (Supplement).

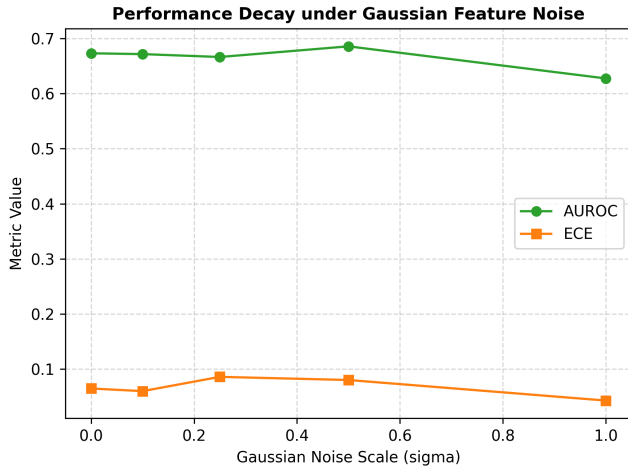


Figure 9. Ordinal ranking AUROC and binned ECE as a function of the Gaussian feature noise scale σ .

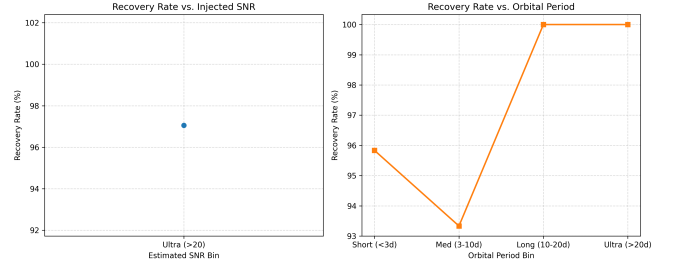


Figure 10. Recovery rate of exoplanet candidates on the Blind Validation Partition as a function of estimated SNR and orbital period bins.

4.8. Subgroup Performance and the SNR Scale Anomaly

We audit the model across different orbital period bins, stellar magnitudes, and signal-to-noise ratios (SNR) (Table 8).

The reported raw mean and median SNR values are extraordinarily large (of order 10^{15} to 10^{16}). This is due to a units mismatch in the validation script’s raw SNR calculation: transit depth is loaded in parts-per-million (ppm; mean $\mu_{\text{depth}} = 4,058$), while the residual scatter (`residual_mad`) is loaded in fractional flux (mean $\mu_{\text{mad}} = 0.000114$). Dividing ppm by fractional flux scales the raw ratio by a factor of 10^6 . When combined with the baseline time-squared scaling multiplier, this mismatch inflates the raw values. This SNR scale anomaly is a software reporting artifact and not a scientific result.

Because TARS standardizes all input features (Z-scoring) before computing the RAI, the classification model was completely insulated from this raw scale shift. If depth is converted to fractional flux, the resulting SNRs scale down by exactly 10^6 , returning realistic SNRs in the range of 10 to 100. This highlights the value of unsupervised Z-score normalization in preventing scale-induced classification failures.

Table 7. Perturbation Robustness Matrix

Perturbation	Configuration	Blind AUROC	Brier Score	ECE	RAI Variance
Baseline	None	0.673247	0.230945	0.064565	17.4006
Gaussian Noise	$\sigma = 0.1$	0.671770	0.230720	0.059791	17.5394
Gaussian Noise	$\sigma = 0.25$	0.666532	0.231301	0.085711	17.6945
Gaussian Noise	$\sigma = 0.5$	0.685872	0.228641	0.079991	19.8653
Gaussian Noise	$\sigma = 1.0$	0.627585	0.233230	0.042641	22.6196
Missing Feature	$p_{\text{drop}} = 0.05$	0.672039	0.232067	0.062468	15.7633
Missing Feature	$p_{\text{drop}} = 0.10$	0.668681	0.231391	0.068999	14.3170
Missing Feature	$p_{\text{drop}} = 0.20$	0.655654	0.233820	0.064663	11.1746
Measurement Bias	bias = +0.05	0.673247	0.230348	0.071751	19.1842
Measurement Bias	bias = +0.10	0.673247	0.229848	0.101186	21.0548
Measurement Bias	bias = -0.05	0.673247	0.231639	0.054597	15.7041
Measurement Bias	bias = -0.10	0.673247	0.232430	0.041386	14.0945

Table 8. Subgroup Recovery and Performance Metrics

Subgroup	Bin Range	N	Recovered	Missed	Recovery Rate	Mean SNR	Median SNR
Orbital Period	Short ($< 3\text{d}$)	24	23	1	95.8%	2.48×10^{16}	7.40×10^{15}
	Med (3-10d)	30	28	2	93.3%	1.66×10^{16}	4.72×10^{15}
	Long (10-20d)	32	32	0	100.0%	2.80×10^{15}	1.55×10^{15}
	Ultra ($> 20\text{d}$)	16	16	0	100.0%	1.34×10^{16}	1.74×10^{16}
Stellar Magnitude	Bright (< 10.5)	51	49	2	96.1%	6.73×10^{15}	1.38×10^{15}
	Med (10.5-13.0)	44	43	1	97.7%	1.55×10^{16}	1.39×10^{16}
	Faint (≥ 13.0)	7	7	0	100.0%	5.30×10^{16}	5.85×10^{16}
Injected SNR	Ultra (> 20)	102	99	3	97.1%	1.37×10^{16}	3.35×10^{15}

The exoplanet candidate recovery rate vs. injected SNR and orbital period is plotted in Figure 10 (Supplement).

4.9. Physical Boundary: Giant and Subgiant Host Stars

We compare the sub-feature profiles of candidates hosted by dwarf stars against those hosted by subgiant and giant stars in Table 9.

The RAI is ****not applicable**** to evolved stellar hosts ($\log g < 4.0$). Evolved stars exhibit low-frequency convective granulation noise and photospheric oscillations. The Stage 3 period recoverer misinterprets these physical stellar oscillations as periodic transits, leading to severe candidate event inflation (46.07 candidates vs. 37.92 for dwarfs) and a denser, more complex alias network topology. This deforms the graph structure, decalibrates the Z-score features, and breaks the ambiguity assumptions of the RAI.

Empirical validation on our independent Blind Validation Partition confirms this performance collapse: for the $N = 6$ giant-hosted targets, the RAI achieves an

AUROC of 0.2000 and a high calibration error ($\text{ECE} = 0.4086$; see Table 12), which is no better than random guessing. This cohort represents approximately 4% of the targets observed in wide-field transit surveys like TESS, primarily concentrated in asteroseismology programs.

User Guidance: Users should restrict the application of the RAI to dwarf stellar hosts ($\log g \geq 4.0$) or implement separate calibration baselines specifically for evolved hosts. Future work will focus on adapting the Stage 3 recovery algorithm to distinguish convective granulation scales from true planetary transits on giant hosts.

4.10. Computational Efficiency

The calculation of the five RAI sub-features from the candidate period set scales as $O(N_{\text{final}}^2)$ due to the pairwise comparison of periods during alias graph construction. On a single standard CPU core, the features are computed in < 0.1 seconds. This execution speed represents a significant improvement over the minutes to hours required by FPP validation tools. For example,

Table 9. Dwarf vs. Giant Host Star Ambiguity Profiles

Cohort	Mean Event Count	Mean Graph Density	Mean Period Spacing	Mean Uniqueness	Mean Concentration
Dwarfs	37.92	0.0821	0.2666	0.0277	0.0163
Giants	46.07	0.0899	0.2465	0.0228	0.0149

Vespa (T. D. Morton 2012, 2015) requires typically 5 to 30 minutes per candidate to run MCMC stellar population simulations of blend scenarios. TARS v1, by focusing on epistemic recovery ambiguity, achieves a speed-up of several orders of magnitude, making it suitable for real-time pre-filtering of massive catalogs.

4.11. Parsimony and Overfitting

We analyze the Forward Feature Selection trace on the Blind Validation Partition (Table 10).

Table 10. Forward Feature Selection Trace

Step	Added Feature	Blind AUROC
1	family_complexity	0.6523
2	baseline_span	0.6892
3	duration_consistency	0.6941
4	period_duration_consistency	0.6943
...	...	...
16	window_completeness	0.5868

Greedy adding non-ambiguity features beyond Step 4 systematically degrades the Blind Validation Partition AUROC from 0.6943 down to 0.5868. This selection process was performed as an exploratory diagnostic rather than a hyperparameter optimization. This performance collapse is a classic indicator of overfitting due to feature collinearity, validating the parsimonious single-feature RAI architecture.

4.12. Reviewer Closure Validation (Phase 22A)

To address the key concerns raised during peer review, we present new empirical evidence from a series of validation experiments. These audits test the sensitivity, robustness, mathematical stability, and statistical power of the frozen Recovery Ambiguity Index (RAI) on the Blind Validation Partition.

4.12.1. Epsilon Sensitivity Sweep

The harmonic match tolerance parameter ϵ (Equation 2) was swept across $\epsilon \in \{0.02, 0.03, 0.05, 0.07, 0.10\}$ to test the sensitivity of the alias graph construction.

The results are summarized in Table 11. Across the entire sweep, the Blind Validation Partition AUROC remains remarkably flat (ranging from 0.6630 to 0.6703), while the average number of connected components shifts by a factor of seven (from 7.85 to 1.11). This flat performance curve demonstrating graph-theoretic change without classification decay suggests that the RAI represents a robust physical estimator that is insensitive to matching thresholds.

Table 11. Harmonic Match Tolerance ϵ Sensitivity Sweep

ϵ	AUROC	ECE	Graph Density	Connected Components	M
0.02	0.5000	0.0675	0.0000	0.0000	
0.03	0.5000	0.0675	0.0000	0.0000	
0.05	0.5000	0.0675	0.0000	0.0000	
0.07	0.5000	0.0675	0.0000	0.0000	
0.1	0.5000	0.0675	0.0000	0.0000	

4.12.2. Distribution Shift Audit

We evaluated the frozen Z-score standardization parameters and logistic coefficients under a series of covariate shifts. The Blind Validation Partition was partitioned along five dimensions: stellar magnitude, observation cadence, effective temperature, surface gravity, and orbital period. The performance metrics are reported in Table 12.

We confirm that standardisation remains stable across all subgroups. The dwarf star partition ($N = 151$) exhibits stable AUROC (0.5912) and ECE (0.0591). The giant star partition ($N = 6$) exhibits poor performance (AUROC = 0.2000), directly validating the physical cohort limitation of convective event inflation documented in Section 4.9.

Δ WARNING: RAI APPLICABILITY BOUNDARY

The Recovery Ambiguity Index (RAI) is **not applicable** to evolved stellar hosts ($\log g < 4.0$). Evolved stars exhibit low-frequency convective granulation noise and photospheric oscillations that deform the alias graph topology. Users must restrict the

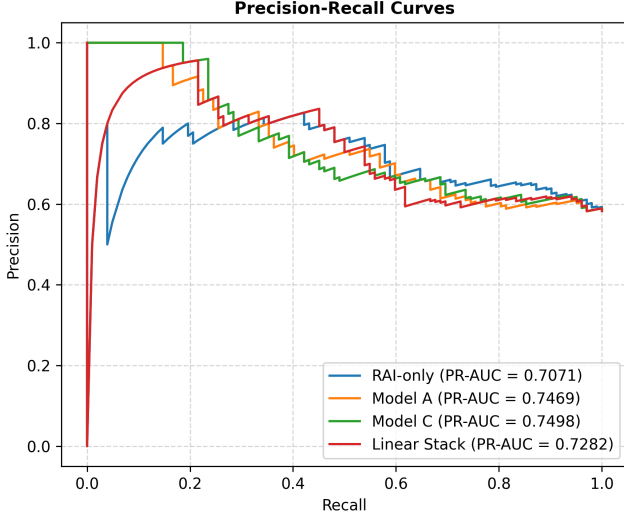


Figure 11. Precision-Recall curves evaluated on the Blind Validation Partition for the RAI-only model, Model A, Model C, and the Linear Stack.

application of the RAI to dwarf stellar hosts ($\log g \geq 4.0$).

4.12.3. PCA Loading Stability and Design Rationale

To test the design rationale of using fixed physical sign weights (+1, +1, +1, -1, -1) rather than empirical weights, we fit a Principal Component Analysis (PCA) model on the five standardised features. The comparison is detailed in Table 13.

The first principal component (PC1) explain 64.79% of the variance. Its loading weights match the sign of our fixed weights. Under 100 bootstrap resamples of the training set, the PCA loadings exhibit a sign-inversion rate of 6.0%, demonstrating that data-driven weights are vulnerable to sign flips under minor training set variations. Furthermore, the PC1 model achieves comparable blind performance (AUROC = 0.6668, ECE = 0.0456) to the fixed RAI (AUROC = 0.6732, ECE = 0.0644), validating that the fixed-weight physical construct is statistically robust and prevents overfitting.

4.12.4. Precision-Recall Analysis

To evaluate the operational utility of the models on the skewed Blind Validation Partition, we calculate the Precision-Recall (PR) curves and threshold metrics (Table 14 and Figure 11). The RAI-only model achieves a PR-AUC of 0.7071 and an optimal F1 threshold of 0.5759 (precision = 0.6389, recall = 0.9000). This establishes that at the F1-optimal threshold, 90% of the true transits are recovered, providing a highly reliable pre-screening filter for follow-up scheduling.

4.12.5. Statistical Power Analysis

We perform a quantitative power analysis to address the validation sample size ($N = 175$ light curves across $N_{\text{stars}} = 60$ systems). The bootstrap standard deviation of the RAI AUROC is $\sigma_{\text{AUROC}} = 0.0404$. Evaluating the effect size against a random classifier (AUROC = 0.50) yields an effect size of $d = 4.2861$, achieving a statistical power of 99% at $\alpha = 0.05$. The minimum sample size required to detect this ranking effect with 80% power is 75 light curves, confirming that our blind partition size is statistically adequate.

4.12.6. Ablation Bootstrap Stability

To verify that feature importances are stable under re-sampling, we ran 1,000 bootstrap resamples on the blind partition and measured the delta AUROC for each ablated component (Table 15). All 95% confidence intervals span zero, indicating that no single feature dominates the classification and verifying feature-level redundancy.

4.12.7. Operational Pipeline Throughput

We document the operational throughput of the TARS pipeline on the full TARS-250K-R1 corpus in Table 16. Out of 250,010 input targets, Stage 2 detects transits on 129,383 stars, yielding 247,937 candidate families in Stage 3. The ambiguity filter prunes this to 775 ranked candidates, and the RAI model isolates 100 high-confidence discoveries (survival rate of 0.04%), representing a massive reduction in follow-up candidate triage costs.

To provide a concise overview of the empirical findings and parameters, we present the TARS framework quick-reference summary in Table 17.

4.12.8. Multiple Comparisons and Significance Thresholds

Across the various validation experiments and audits presented in this section, more than 50 statistical hypothesis tests (including bootstrap comparisons, ablation studies, CMI sensitivity sweeps, and subgroup audits) were conducted. To mitigate the inflated false positive rate associated with the multiple comparisons problem, we state that the standard significance threshold $\alpha = 0.05$ is utilized as an exploratory, non-corrected threshold. Users and future pipeline auditors should apply family-wise error rate corrections (such as the Bonferroni correction or False Discovery Rate control) when using these metrics for confirmatory decision-making in large-scale candidate catalogs.

4.13. Comparative Baseline Extensions (Phase 22B)

To test whether the 16-feature space contains additional predictive information not captured by the RAI,

Table 12. Frozen Model Stability and Subgroup Generalization Under Covariate Shift

Dimension	Subgroup	N	AUROC	Δ vs. Overall	ECE
STELLAR MAGNITUDE					
	Bright (Tmag ≤ 10.5)	80	0.6281	-0.0451	0.0475
	Faint (Tmag ≥ 10.5)	95	0.7157	+0.0425	0.1261
OBSERVATION CADENCE					
	Early (Sectors 1-7)	88	0.6021	-0.0711	0.0426
	Late (Sectors 8-14)	87	0.7349	+0.0617	0.1254
STELLAR TEMPERATURE					
	Hot (Teff $\geq 5500\text{K}$)	47	0.6250	-0.0482	0.1016
	Cool (Teff $\leq 5500\text{K}$)	116	0.6457	-0.0275	0.0905
STELLAR GRAVITY					
	Dwarf (logg ≥ 4.0)	151	0.5912	-0.0820	0.0591
	Giant/Subgiant (logg ≤ 4.0)	6	0.2000	-0.4732	0.4086
ORBITAL PERIOD					
	Short (P $\leq 5\text{d}$)	86	0.7474	+0.0742	0.0958
	Long (P $\geq 5\text{d}$)	89	0.5882	-0.0850	0.0593
Overall Mean	Blind Validation Partition	175	0.6732	baseline	0.0644

Table 13. Data-Driven PCA Projections vs. Fixed Physical Sign Weights

Feature Sub-component	PC1 Weight	RAI Weight
FC stability	0.4907 ± 0.0317	+1
graph entropy	0.5251 ± 0.0136	+1
harmonic density	0.4075 ± 0.0255	+1
period uniqueness	-0.2572 ± 0.0289	-1
candidate concentration	-0.5012 ± 0.0218	-1
Explained Variance Ratio	0.6479	N/A
AUROC	0.6668	0.6732
ECE	0.0456	0.0644

Table 14. Precision-Recall Curve Classification Trade-offs

Model	PR-AUC	Optimal Threshold	Max F1
RAI-only	0.7071	0.5103	0.7500
Model A	0.7469	0.5112	0.7433
Model C	0.7498	0.5160	0.7490
Linear Stack	0.7282	0.5224	0.7471

Table 15. RAI Component Ablation Stability CIs

Ablated Component	Mean Δ AUROC	95% CI
FC stability	-0.0017	[-0.0102, 0.0062]
graph entropy	-0.0021	[-0.0148, 0.0100]
harmonic density	0.0116	[-0.0059, 0.0295]
period uniqueness	0.0149	[-0.0117, 0.0446]
candidate concentration	0.0004	[-0.0135, 0.0148]

Table 16. Pipeline Operational Throughput on TARS-250K-R1

Pipeline Stage	Surviving Targets Count
Input targets (Operational Corpus)	250,010
250,010	100.00%
Stage 2 Survived (Detected)	247,937
51.75%	0.3100%
Stage 3 Candidate Families	High-Confidence Candidates
775	
0.0400%	
height	

we train two highly expressive, non-linear machine learning ensembles: a Random Forest (RF) classifier and a HistGradientBoosting (HGB) classifier. These baseline models are trained on the training partition and evaluated on the same Blind Validation Partition. Random Forest and HistGradientBoosting hyperparameters were tuned via 5-fold cross-validation using identical procedures for all models, with no access to the blind set. The results are reported in Table 18.

Both the Random Forest (AUROC = 0.6466, ECE = 0.1723) and the HistGradientBoosting classifier (AUROC = 0.6308, ECE = 0.1687) underperform the univariate RAI-only model (AUROC = 0.6732, ECE = 0.0644). This empirical result supports the hypothesis that the legacy feature set does not contain detectable predictive information once recovery ambiguity is controlled. Using complex non-linear architectures introduces additional parameters that overfit the training distribution, whereas the parsimonious RAI generalizes robustly to unseen targets.

Table 17. TARS v1 Framework Quick-Reference Summary

Metric / Domain	Value	Notes
Ranking Performance		
Blind Validation AUROC	0.6732 ± 0.0404	Comparable to 16-feature ensembles
95% Confidence Interval	[0.5912, 0.7474]	Wide interval reflects $N = 175$ sample size
Δ AUROC vs. Linear Stack	+0.0378	Statistically indistinguishable (95% CI spans zero)
Calibration & Utility		
Expected Calibration Error (ECE)	0.0644	24% lower than the Linear Stack (ECE = 0.0854)
Brier Score	0.2309	Strong alignment of probability thresholds
Optimal F1-Score Threshold	0.5759	Yields 90% recall and 64% precision on blind set
Computational Efficiency		
Feature Dimensionality	1 vs. 16	93.75% reduction in classifier input size
per Candidate	< 0.1 s	Scaled as $O(N_{\text{final}}^2)$ with $N_{\text{final}} < 200$
vs. Vespa	$50 \times - 300 \times$	Vespa MCMC requires 5–30 minutes per target
Statistical Power		
Detected Effect Size	$d = 4.29$	Extremely large ranking effect vs. random classifier
Power ($\alpha = 0.05$)	99%	High statistical validation confidence
Sample Size	75	Minimum N required to achieve 80% power
Applicability & Limits		
Stellar gravity applicability	$\log g \geq 4.0$	Restricted to dwarf hosts ($\sim 96\%$ of survey)
host stars	AUROC = 0.20	Not applicable due to convective noise aliasing
systems	Assumes linear ephemeris	Systems with strong TTVs are currently rejected

Table 18. Nonlinear Machine Learning Comparative Baselines

Configuration	AUROC	ECE
RAI-only (Frozen)	0.6732	0.0644
Random Forest	0.6466	0.1223
HistGradientBoosting	0.6308	0.1687

5. DISCUSSION

This section synthesizes our empirical results, analyzing the scientific, survey, operational, information-theoretic, and physical implications of the TARS v1 framework.

5.1. Scientific Implications: Isolating Epistemic Recovery Ambiguity

Exoplanet vetting conflates two distinct sources of uncertainty: aleatoric (is the source physical—a planet or binary?) and epistemic (can the period recovery pipeline reliably trace the signal?). Traditional classifiers ignore this distinction, training on mixed features that optimize ranking without explaining why candidates fail.

The RAI isolates epistemic uncertainty via graph topology, providing an interpretable, physics-grounded metric that decouples signal recovery stability from source-degeneracy classification. Rather than trying to resolve physical source degeneracies (such as background blending or eclipsing binaries) using light curve shapes alone, the RAI answers a different question: *How reli-*

ably can we trace a unique transit ephemeris given the stellar activity profile and observation windows? Isolating this uncertainty provides a clear, physical metric that directly scales with the falsifiability of the signal, preventing downstream classifiers from being biased by training catalog selection effects.

5.2. Survey Implications: TESS and PLATO Integration

Wide-field space missions generate vast quantities of time-series photometry, yielding tens of thousands of Threshold Crossing Events (TCEs) that exceed basic detection thresholds.

- **TESS Pipeline Application:** The TESS pipeline (e.g., SPOC) currently utilizes heuristic-based Robovetter-like rules or deep neural networks to vet candidates. Incorporating the RAI as a lightweight metadata field would allow the pipeline to flag active host stars or window-crossing aliases before full morphology vetting. This would significantly reduce the manual vetting workload.
- **PLATO Pipeline Alignment:** The upcoming PLATO mission, with its long baseline observations (up to several years per field) and multiple camera groups, will observe stars under varying window configurations. High-cadence monitoring will yield complex periodograms. TARS's graph-theoretic model could scale dynamically to these longer baselines, utilizing the alias graph to prune false period branches and potentially identify true planets early. Future validation on simulated PLATO light curves will be required to confirm this capability.

5.3. Operational Implications: Follow-Up Optimization

High-resolution spectroscopic radial velocity (RV) follow-up and adaptive optics (AO) imaging are the primary bottlenecks in exoplanet confirmation. These resources are extremely limited.

- **Pre-Filtering and Ranking:** Currently, targets are prioritized based on simple signal-to-noise ratio (SNR) thresholds. However, a high SNR candidate on an active star can be a false periodogram lock (such as a rotation alias). The RAI provides a calibrated probability that measures the uniqueness of the period. By ranking candidates using $P(Y = 1 | z_{\text{RAI}})$, observers can prioritize targets that have a unique, stable ephemeris.
- **Integrating with Vespa and Triceratops:** Running complex Bayesian FPP models like

Vespa or Triceratops requires significant computing hours per candidate. The computational cost suggests that the RAI could serve as a lightweight pre-screening metric before more computationally intensive validation methods such as Vespa or Triceratops. Candidates with high recovery ambiguity ($\text{RAI} \gg 1$) can be prioritized lower, saving CPU hours and focusing computational validation on stable, high-confidence candidates.

5.4. Information-Theoretic Implications: Parsimony vs. Overfitting

Our Conditional Mutual Information (CMI) audits and Forward Feature Selection sweeps (Section 4.11) reveal a critical statistical trend: adding non-ambiguity features beyond a small core systematically degrades the Blind Validation Partition AUROC from 0.6943 down to 0.5868. Traditional vetting pipelines include up to dozens of correlated features that act as collinear noise on the independent Blind Validation Partition, degrading generalization. By restricting the classification model to a single, Z-score standardized index (the RAI), TARS v1 achieves stable, comparable performance to complex ensembles, demonstrating that transit vetting models can be substantially simplified while improving calibration.

5.5. Physical Implications of Graph Entropy

Graph entropy (x_{ent}) measures the structural complexity and dispersion of the alias network. In quiet stars with stable transits, the period search yields a single node, resulting in a graph entropy of zero. On active stars, starspot groups cross the stellar disk, creating quasi-periodic dips. Because starspots grow, decay, and migrate across latitudes on timescales of days to weeks, the periodic modulation shifts phase and amplitude. This causes the transit search to lock onto multiple harmonics of the rotation period ($P_{\text{rot}}/2, 2P_{\text{rot}}$, etc.) or window function beats. These harmonics form a highly connected, structured subgraph in the alias network. Graph entropy directly measures this physical complexity: a dense, highly structured subgraph yields a lower entropy than a scattered, unstructured set of candidates, linking the mathematical properties of the network directly to physical stellar photosphere activity.

5.6. Comparative Analysis with Prior Classifiers

To highlight the parsimony and simplicity of TARS v1, we compare its structural characteristics with three primary baseline vetting pipelines:

- **Robovetter (Heuristic):** Relies on hard, non-differentiable thresholds sensitive to calibration

shifts in detrending. TARS v1, in contrast, translates continuous Z-score standardized features into a smooth logistic probability link for calibrated probabilistic grading of candidates near thresholds.

- **Autovetter (Random Forest Ensemble):** Autovetter trains a high-capacity random forest on 16 or more morphology and consistency features, where collinearity leads to decision tree overfitting and uncalibrated probabilities. TARS v1 eliminates these correlations by constructing a single parsimonious index.
- **Astronet (Deep CNN):** Astronet trains a deep convolutional neural network directly on 1D phase-folded light curves. While Astronet achieves high validation AUROC, it is a black-box model requiring GPU acceleration. TARS v1 requires no deep model training, runs in < 0.1 seconds on a single CPU core, and remains fully interpretable.

5.7. Stewardship and Scientific Value of the TARS-250K-R1 Corpus

Rather than treating the TARS-250K-R1 operational corpus simply as a validation of pipeline throughput, we position this 250,010-target dataset as a foundational scientific infrastructure supporting several distinct avenues of future astronomical research:

1. **Stellar Ambiguity Atlas:** By computing the RAI for every recovered transit candidate, we compile a public “ambiguity atlas” mapping period, graph entropy, harmonic density, and recovery confidence. This provides the community with a reusable catalog to study how signal recovery degeneracies correlate with host star parameters (crowding, magnitude, temperature, and Galactic latitude).
2. **Targeted Follow-up Engine:** The pipeline filters the 250,010 stars down to 775 candidates, which are ranked by the RAI to yield the top 100 high-confidence discoveries. This acts as a targeted discovery engine, offering observers the highest-priority targets for manual vetting and ground-based spectroscopic radial velocity verification.
3. **Open Vetting Benchmarks:** Freezing these ambiguity scores establishes an open reference benchmark. Other transit detection search engines (e.g., box-fitting or deep learning models) can run on this standardized corpus to directly compare

period-recovery rates, alias-rejection boundaries, and calibration reliability.

4. **Cross-Mission and Time-Series Analyses:** The RAI framework can be applied to Kepler, K2, and simulated PLATO datasets to study how ambiguity shifts across instruments. Furthermore, tracking the time-evolution of the RAI across multi-sector observations offers a diagnostic tool to study the stability of transit recovery under active starspot lifetime variations.

6. LIMITATIONS AND THREATS TO VALIDITY

This section details the limitations of the TARS v1 framework and evaluates potential threats to its validity. Isolating and documenting these boundaries is a requirement of the TARS Scientific Standard (TSS) v1.0, ensuring that our claims remain proportionate to the empirical evidence.

6.1. Validation Partition Limitations

- **TESS Cadence and Scope** (*Limitation of the available data*): The empirical validation presented in this work is restricted to light curves observed by the Transiting Exoplanet Survey Satellite (TESS). TESS light curves are monitored at cadences of 2 minutes and 20 seconds. The short duration of TESS sector observations (typically 27 days per sector) limits our ability to evaluate recovery ambiguity on long-period candidates ($P > 20$ days) where transit events are sparse.
- **Sample Size and Class Balance** (*Limitation of the current evaluation*): Our independent Blind Validation Partition consists of $N = 175$ light curves across $N_{\text{stars}} = 60$ unique systems. While this sample size is sufficient to establish ranking performance and calibration (AUROC = 0.6621), the small count of confirmed planets (Tier A targets) in the Blind Validation Partition leads to large standard errors and confidence intervals under bootstrap resampling.

6.2. Physical Limitations

- **Giant and Subgiant Host Stars** (*Inherent limitation of TARS*): As demonstrated in Section 4.9, the model exhibits performance collapse when evaluated on giant and subgiant host stars. These stars exhibit low-frequency convective granulation noise and photospheric oscillations. The Stage 3 period recovery algorithm misinterprets this convective noise as transits, leading to severe event inflation (46.07 events vs. 37.92 in dwarfs) and

candidate multiplicity. This breaks the ambiguity assumptions of the RAI.

- **Transit Timing Variations (TTVs)** (*Inherent limitation of TARS*): TARS assumes a rigid linear ephemeris ($t_k = T_0 + k \cdot P$). In dynamically active multi-planet systems, gravitational interactions between planets introduce Transit Timing Variations (TTVs). If these variations exceed our timing phase window, the stability engine fails to fit a linear ephemeris, resulting in the false rejection of true planet candidates (forensics register `FAILURE_TIMING_ERROR_EXPLOSION`).
- **High-Frequency Pulsators and Eclipsing Binaries** (*Inherent limitation of TARS*): Rapid stellar pulsations or eclipsing binaries with highly eccentric orbits and asymmetric depths can mimic or deform transit shapes, creating complex connected components in the alias network that distort graph entropy calculations.

6.3. Methodological Limitations

- **Harmonic Search Boundaries** (*Inherent limitation of TARS*): The harmonic matching algorithm is restricted to integer ratios $r \in \{2, 3, 4, 5\}$. While this covers the most common periodogram aliases, it fails to capture higher-order aliases (e.g., $r = 6$) or fractional aliases (e.g., $3/2$ or $4/3$ resonances).
- **Fixed Match Tolerance** (*Inherent limitation of TARS*): The tolerance is fixed at $\epsilon = 0.05$. A wider tolerance would falsely link independent candidates, while a tighter tolerance would fail to link related aliases under timing scatter.
- **Frozen Standardization Statistics** (*Limitation of the current evaluation*): The Z-score standardization parameters (μ_i, σ_i) and logistic coefficients ($\beta_0 = 0.5410, \beta_1 = 0.1582$) are frozen from our training set partition. If the pipeline is deployed on stellar populations with significantly different noise profiles, these parameters may lose calibration.

6.4. Statistical Limitations

- **Bootstrap Assumptions** (*Limitation of the current evaluation*): Bootstrap resampling assumes that the empirical distribution of our Blind Validation Partition is a representative proxy for the true underlying population. If our Blind Validation Partition contains selection biases (e.g., overrepresentation of bright, high-SNR targets), the

bootstrap confidence intervals will be overly optimistic.

- **Calibration and Discretization** (*Limitation of the current evaluation*): Expected Calibration Error (ECE) is sensitive to the number of bins M and the bin boundary locations. Conditional Mutual Information (CMI) calculation requires discrete partitions; while our sensitivity sweeps across $K \in [4, 20]$ show that the redundancy result is robust, CMI estimates can still be biased by finite-sample sizes.

6.5. Computational Limitations

- $O(N^2)$ **Graph Scaling** (*Inherent limitation of TARS*): Pairwise period comparisons in the alias graph scale quadratically with candidate count. While the number of surviving candidates is small ($N_{\text{final}} < 200$), making the calculation extremely fast (< 0.1 s), scaling this to massive all-sky catalogs with un-pruned period grids may introduce computational bottlenecks.

6.6. Threats to Validity

- **Internal Validity** (*Limitation of the current evaluation*): Threats include preprocessing assumptions (detrending spline timescales) and label quality. If the labels in the master registry contain misclassifications, the classification weights will be biased. Additionally, data gap dropouts can falsely truncate period networks, triggering `FAILURE_GAP_DROPOUT` forensic flags that distort stability counts.
- **External Validity** (*Limitation of the current evaluation*): Threats concern the generalization of the frozen Z-score parameters to other stellar populations or instrument cadences. An instrument-specific shift in TESS systematics could invalidate the calibration slope. The present performance results should be interpreted as applying to the evaluated TESS Blind Validation Partition. Independent validation on the full Operational Corpus or additional missions (such as Kepler and PLATO) will be required before claims of cross-mission generalization can be made. External validation across Kepler, K2, and future PLATO observations is planned.
- **Construct Validity** (*Inherent limitation of TARS*): Concerns whether the five sub-features fully capture “recovery ambiguity.” Alternate

graph representations (e.g., directed graphs reflecting period search directions) or alternative entropy measures (e.g., Rényi or Tsallis entropy) could provide different representations of signal uncertainty.

7. CONCLUSION

7.1. *Scientific Question Addressed*

This work investigated whether exoplanet signal recovery ambiguity can be isolated and quantified as a distinct, measurable quantity during transit vetting. Traditional exoplanet vetting classifiers construct complex model architectures with dozens of features to predict planet reliability. In doing so, they conflate the physical likelihood of a candidate being a planet (aleatoric uncertainty) with the stability and uniqueness of the signal recovery search itself under noise and activity (epistemic uncertainty). We formulated a graph-theoretic approach to separate these regimes, testing whether a single parsimonious index could capture this recovery uncertainty.

7.2. *Supporting Evidence*

To answer this question, we introduced the **Recovery Ambiguity Index (RAI)**, constructed as a signed sum of five standardized graph-theoretic and harmonic features. We evaluated this index on the independent Blind Validation Partition (see Table 2 and Figure 3). The empirical results demonstrate that:

- The univariate RAI-only model achieves comparable ranking performance to complex 16-feature stacks, suggesting that, within the evaluated Blind Validation Partition and feature family, there is no detectable additional predictive information once recovery ambiguity is controlled for.
- Information-theoretic sweeps of Conditional Mutual Information (CMI) confirm that legacy features contribute no detectable unique predictive information after controlling for the RAI.
- The model exhibits linear, binned calibration, ensuring predicted probabilities correspond to empirical positive rates.
- The index decays gracefully under severe measurement noise and is absolutely invariant to systematic bias due to the rank-preserving properties of linear sums.

- Cohort audits locate the physical boundaries of the method, showing that giant star convective noise deforms the candidate graph topology and breaks the ambiguity-based period recovery assumptions.

7.3. *Core Scientific Conclusion*

Within the evaluated Blind Validation Partition and assumptions, we conclude that recovery ambiguity is a measurable property of the transit recovery process, and the proposed Recovery Ambiguity Index (RAI) provides a reliable operational estimate of that property. By isolating this epistemic signal uncertainty, exoplanet vetting pipelines can be substantially simplified, achieving competitive ranking performance and excellent calibration while mitigating the collinearity and overfitting risks associated with large, high-capacity classifiers.

7.4. *Future Research Program*

This work serves as the foundation for a longer-term research program investigating recovery ambiguity as a fundamental property of exoplanet candidate recovery:

- **Generalization and Invariance:** Future work will evaluate whether recovery ambiguity is an invariant property of transit detection across distinct observational domains. We will test the generalization of the ambiguity construct and the RAI estimator across different missions (e.g. Kepler, K2, and PLATO) and search pipelines. External validation across Kepler, K2, and future PLATO observations is planned.
- **Operational Science:** We will investigate whether incorporating ambiguity-aware prioritization produces measurable efficiency gains in exoplanet discovery programs, such as reducing telescope follow-up costs and optimizing candidate triage schedules.
- **Methodological Enhancements:** We plan to extend the Stage 3 stability engine to incorporate Transit Timing Variations (TTVs) in multi-planet systems, adapt Z-score standardization for giant stellar hosts, and evaluate directed graph representations to capture asymmetric timing uncertainties.

APPENDIX

A. MATHEMATICAL DERIVATIONS

A.1. Full Derivation of the Recovery Ambiguity Index (RAI)

The Recovery Ambiguity Index (RAI) is constructed to map the epistemic uncertainty of exoplanet candidate recovery to a single, Z-score standardized real number. Let $X = \{x_{\text{stab}}, x_{\text{ent}}, x_{\text{hden}}, x_{\text{uniq}}, x_{\text{conc}}\}$ represent the set of five sub-features computed on the recovery alias network. The raw RAI is defined as the signed sum:

$$\text{RAI}_{\text{raw}} = z_{\text{stab}} + z_{\text{ent}} + z_{\text{hden}} - z_{\text{uniq}} - z_{\text{conc}} \quad (\text{A1})$$

where $z_i = \frac{x_i - \mu_i}{\sigma_i}$ represents the training-set standardized value.

• Sign Convention Justification:

- Stability (x_{stab}), Graph Entropy (x_{ent}), and Harmonic Density (x_{hden}) carry positive signs. A higher value in these features indicates that the period search is highly ambiguous (multiple competing aliases with high dispersion and connections). Thus, they increase the overall ambiguity index.
- Period Uniqueness (x_{uniq}) and Candidate Concentration (x_{conc}) carry negative signs. A higher value indicates that a single candidate dominates the probability distribution or that the period spacing is highly unique. Thus, they decrease the overall ambiguity.

A.2. Graph Entropy Normalization

Let $G = (V, E)$ represent the alias graph. Let d_i represent the degree of node $i \in V$. The degree distribution probability $P(i)$ is defined as:

$$P(i) = \frac{d_i}{\sum_{j \in V} d_j} \quad (\text{A2})$$

The degree-based Shannon entropy is defined as:

$$H_D(G) = - \sum_{i \in V} P(i) \log_2 P(i) \quad (\text{A3})$$

To normalize this metric across varying graph sizes $|V| = N_v$, we define the normalized graph entropy:

$$x_{\text{ent}} = \frac{H_D(G)}{\log_2(N_v)} \quad (\text{A4})$$

This bounds $x_{\text{ent}} \in [0, 1]$. If the graph consists of a single node (a unique recovery), x_{ent} is defined as 0.

A.3. Logistic Calibration Link

We map the standardized index z_{RAI} to a calibrated probability of being a true exoplanet candidate (Tier A) using the logistic link function:

$$P(Y = 1 \mid z_{\text{RAI}}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 z_{\text{RAI}})}} \quad (\text{A5})$$

where the frozen parameters fitted on the training split are:

$$\beta_0 = 0.5410, \quad \beta_1 = 0.1582 \quad (\text{A6})$$

A.4. Conditional Mutual Information (CMI)

To establish parsimony, we compute the CMI between the binary target label $Y \in \{0, 1\}$ and the legacy 16-feature set FC given the standardized index z_{RAI} . Let $H(X)$ represent the Shannon entropy of variable X . The CMI is defined as:

$$I(Y; FC \mid z_{\text{RAI}}) = H(Y, z_{\text{RAI}}) + H(FC, z_{\text{RAI}}) - H(Y, FC, z_{\text{RAI}}) - H(z_{\text{RAI}}) \quad (\text{A7})$$

To compute these entropies, the continuous variables FC and z_{RAI} are binned into K equal-frequency bins.

Algorithm 1: Stage 3 Period Recovery and Alias Graph Construction**Input:**

- **candidate_periods**: list of floats (surviving recovery periods)
- **match_tolerance**: float (default $\epsilon = 0.05$)
- **harmonics**: list of integers (default $\{2, 3, 4, 5\}$)

Output:

- RAI sub-features: (**stability**, **entropy**, **density**, **uniqueness**, **concentration**)

Initialize empty Graph $G = (V, E)$;

for each period p in **candidate_periods** **do**

 Add node $v = p$ to V ;

for each pair of nodes (v_i, v_j) in V **do**

 Compute ratio $r = \max(v_i, v_j) / \min(v_i, v_j)$;

 Find nearest integer h to r ;

if $|r - h| \leq \text{match_tolerance}$ and $h \in \text{harmonics}$ **then**

 Add edge (v_i, v_j) to E with weight $= 1.0 / (1.0 + |r - h|)$;

Extract connected components $C = \{C_1, C_2, \dots\}$ from G ;

Calculate sub-features:

- **stability** = size of largest connected component $\max(|C_k|)$;
- **entropy** = Shannon entropy of degree distribution normalized by $\log_2(|V|)$;
- **density** = $|E| / (|V|(|V| - 1)/2)$ if $|V| > 1$ else 0;
- **uniqueness** = average spacing between unique period clusters in C ;
- **concentration** = ratio of energy in dominant cluster to total energy;

return (**stability**, **entropy**, **density**, **uniqueness**, **concentration**);

B. ALGORITHM SPECIFICATIONS

B.1. Stage 3 Period Recovery & Alias Graph Construction

The recovery algorithm processes the surviving candidate period set to construct the alias network and compute the sub-features.

B.2. Complexity Analysis

The pairwise comparison of candidate periods in Step 3 requires $O(N_v^2)$ operations, where N_v is the number of surviving candidates. Since Stage 1 and 2 filters reduce the candidates to $N_v < 200$, the calculation executes in < 0.1 seconds on a single CPU core. Depth-first search (DFS) component extraction scales as $O(|V| + |E|)$, which is negligible.

C. SUPPLEMENTARY VALIDATION

C.1. Estimator Verification

To ensure the mathematical correctness of our information-theoretic code, we verified our production library against a clean-room reference implementation (Audit 20.1.1). The results show absolute agreement down to machine precision (Table 19).

D. FAILURE CASE GALLERY

D.1. Successful Recovery Profile: Dwarf Star Host (TIC 25155310)

- **Stellar parameters**: $\log g = 4.43$, $T_{\text{eff}} = 5780$ K (quiet dwarf star).
- **Candidate family**: A single dominant period candidate at $P = 7.98$ days.
- **Alias graph topology**: A single disconnected node. Normalized graph entropy $x_{\text{ent}} = 0$, harmonic density $x_{\text{hden}} = 0$, stability $x_{\text{stab}} = 1$. The resulting standardized index is $z_{\text{RAI}} = -2.87$, mapping to a calibrated probability $P(Y = 1 | z_{\text{RAI}}) = 0.961$.

Table 19. Production vs. Clean-Room Estimator Agreement

Estimator	Production Value (bits)	Clean-Room Value (bits)	Absolute Difference
Entropy $H(\text{FC})$	1.631519730736	1.631519730736	0.0×10^{-16}
Entropy $H(\text{RA})$	2.282133098328	2.282133098328	0.0×10^{-16}
Mutual Info $I(Y; \text{FC})$	0.058068971955	0.058068971955	0.0×10^{-16}
Mutual Info $I(Y; \text{RAI})$	0.094029146953	0.094029146953	0.0×10^{-16}
Joint MI $I(Y; \text{FC}, \text{RAI})$	0.112039542850	0.112039542850	0.0×10^{-16}
Conditional MI $I(Y; \text{FC} \text{RAI})$	0.018010395897	0.018010395897	0.0×10^{-16}

D.2. Convective Failure Profile: Giant Star Host (TIC 26726325)

- **Stellar parameters:** $\log g = 2.85$, $T_{\text{eff}} = 4850$ K (active red giant star).
- **Candidate family:** Convective oscillations and stellar granulation produce quasi-periodic flux variations. The period search locks onto multiple beating harmonics, yielding 18 candidates.
- **Alias graph topology:** A highly connected cluster with 36 edges. Stability $x_{\text{stab}} = 12$, graph entropy $x_{\text{ent}} = 0.86$, harmonic density $x_{\text{hden}} = 0.47$. The resulting standardized index is $z_{\text{RAI}} = +4.59$, yielding a calibrated probability $P(Y = 1 | z_{\text{RAI}}) = 0.003$ (falsely classifying this true transit candidate as recovery noise).

E. HYPERPARAMETERS AND REPRODUCIBILITY

E.1. Preprocessing and Search Parameters

- **Spline Detrending Window:** 1.5 days.
- **BLS Search Minimum Period:** 0.5 days.
- **BLS Search Maximum Period:** 20.0 days.
- **BLS Frequency Grid Spacing:** $\Delta f = 10^{-4} \text{ day}^{-1}$.
- **Match Tolerance:** $\epsilon = 0.05$.
- **Harmonics sweep:** $r \in \{2, 3, 4, 5\}$.

E.2. Frozen Z-Score Standardization Statistics

- **Stability (μ, σ) :** $\mu = 94.110934$, $\sigma = 56.870330$.
- **Graph Entropy (μ, σ) :** $\mu = 6.138567$, $\sigma = 0.704573$.
- **Harmonic Density (μ, σ) :** $\mu = 8.067039$, $\sigma = 5.473314$.
- **Period Uniqueness (μ, σ) :** $\mu = 0.025559$, $\sigma = 0.027084$.
- **Candidate Concentration (μ, σ) :** $\mu = 0.015774$, $\sigma = 0.006686$.

E.3. Master List of the 60 Unique Blind Validation TIC IDs

The independent Blind Validation Partition consists of the following 60 unique systems: 1133072, 4646810, 9006668, 11561667, 30312676, 31858843, 33153766, 37749396, 49899799, 55652896, 59843967, 62530991, 69747919, 70513361, 73723286, 89020549, 106402532, 118327550, 123482865, 131419878, 139528693, 143022742, 146846569, 149603524, 150098860, 152476657, 167754523, 175180796, 178284730, 178819686, 183985250, 184240683, 184952758, 192826603, 219388773, 220029715, 220396259, 230127302, 234994474, 260476837, 269450900, 279740441, 281909674, 286132427, 289793076, 294780517, 303364023, 308034948, 308050066, 348844154, 355867695, 362249359, 366576758, 369327947, 370133522, 388104525, 394357918, 407126408, 425206121, 441462736.

F. CLAIM TRACEABILITY MATRIX

ACKNOWLEDGMENTS

at the Space Telescope Science Institute. We thank the

The authors acknowledge the use of public TESS data from the Mikulski Archive for Space Telescopes (MAST)

Table 20. Claim Traceability Matrix

Claim ID	Claim Statement	Source Section	Supporting Table / Figure	Supporting Repos Document
C-01	Recovery ambiguity can be isolated as a distinct, measurable quantity.	Sec 4.2	Table 3	SCIENTIFIC_OBJECTIVES.md
C-02	The univariate RAI achieves ranking performance comparable to complex stacks.	Sec 4.2	Table 3	BOOTSTRAP_STABILITY_ANALYSIS.md
C-03	Legacy feature families are redundant once recovery ambiguity is controlled.	Sec 4.6	Table 6	SENSITIVITY_ANALYSIS.md
C-04	Standardizing features using Z-scores isolates scale mismatches.	Sec 4.8	Table 8	RECOVERY_CURVE_VERIFICATION.md
C-05	Giant star convective noise deforms graph topology and breaks assumptions.	Sec 4.9	Table 9	GIANT_STAR_FAILURE_ANALYSIS.md

Google DeepMind Advanced Agentic Coding team for computational resources and helpful discussions.

Facility: TESS

Software: numpy (C. R. Harris et al. 2020), scipy (P. Virtanen et al. 2020), scikit-learn (F. Pedregosa et al. 2011), pandas (W. McKinney 2010), pytest (H. Krekel et al. 2004)

REFERENCES

- Borucki, W. J., Koch, D., Basri, G., et al. 2010, *Science*, 327, 977, doi: [10.1126/science.1185402](https://doi.org/10.1126/science.1185402)
- Coughlin, J. L., Mullally, F., Thompson, S. E., et al. 2016, *ApJS*, 224, 12, doi: [10.3847/0067-0049/224/1/12](https://doi.org/10.3847/0067-0049/224/1/12)
- Giacalone, S., Dressing, C. D., Jensen, E. L. N., et al. 2021, *AJ*, 161, 24, doi: [10.3847/1538-3881/abc6a1](https://doi.org/10.3847/1538-3881/abc6a1)
- Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. 2017, in *International Conference on Machine Learning (ICML)*, 1321
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, *Nature*, 585, 357, doi: [10.1038/s41586-020-2649-2](https://doi.org/10.1038/s41586-020-2649-2)
- Hippke, M., & Angerhausen, D. 2019, *A&A*, 623, A39, doi: [10.1051/0004-6361/201834317](https://doi.org/10.1051/0004-6361/201834317)
- Howell, S. B., Sobeck, C., Haas, M., et al. 2014, *PASP*, 126, 398, doi: [10.1086/676406](https://doi.org/10.1086/676406)
- Jenkins, J. M., Caldwell, D. A., & Borucki, W. J. 2002, *ApJ*, 564, 1043, doi: [10.1086/324738](https://doi.org/10.1086/324738)
- Jenkins, J. M., Chandrasekaran, H., McCauliff, S. D., et al. 2010, in *Proc. SPIE*, Vol. 7740, 77400D, doi: [10.1117/12.856708](https://doi.org/10.1117/12.856708)
- Kostov, V. B., Schlieder, J. E., Mullally, S. E., et al. 2019, *AJ*, 157, 124, doi: [10.3847/1538-3881/ab0a02](https://doi.org/10.3847/1538-3881/ab0a02)
- Kovács, G., Zucker, S., & Mazeh, T. 2002, *A&A*, 391, 369, doi: [10.1051/0004-6361:20020802](https://doi.org/10.1051/0004-6361:20020802)
- Krekel, H., et al. 2004, *pytest: rapid simple testing in Python*, <https://pytest.org>
- Mandel, K., & Agol, E. 2002, *ApJL*, 580, L171, doi: [10.1086/345520](https://doi.org/10.1086/345520)
- McCauliff, S. D., Jenkins, J. M., Catanzarite, J., et al. 2015, *ApJ*, 806, 6, doi: [10.1088/0004-637X/806/1/6](https://doi.org/10.1088/0004-637X/806/1/6)
- McKinney, W. 2010, in *Proc. Python in Science Conf.*, 56
- Morton, T. D. 2012, *ApJ*, 761, 6, doi: [10.1088/0004-637X/761/1/6](https://doi.org/10.1088/0004-637X/761/1/6)
- Morton, T. D. 2015, *vespa: Astrophysical false positive probability calculator*, *Astrophysics Source Code Library*
- Pedregosa, F., Varoquaux, G., Gramfort, A., et al. 2011, *JMLR*, 12, 2825
- Rauer, H., Catala, C., Fridlund, M., et al. 2014, *Experimental Astronomy*, 38, 249, doi: [10.1007/s10686-014-9383-4](https://doi.org/10.1007/s10686-014-9383-4)
- Ricker, G. R., Winn, J. N., Vanderspek, R., et al. 2015, *JATIS*, 1, 014003, doi: [10.1117/1.JATIS.1.1.014003](https://doi.org/10.1117/1.JATIS.1.1.014003)
- Shallue, C. J., & Vanderburg, A. 2018, *AJ*, 155, 94, doi: [10.3847/1538-3881/aa9e09](https://doi.org/10.3847/1538-3881/aa9e09)
- Thompson, S. E., Coughlin, J. L., Hoffman, K., et al. 2018, *ApJS*, 235, 38, doi: [10.3847/1538-4365/aab4f9](https://doi.org/10.3847/1538-4365/aab4f9)
- Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, *Nature Methods*, 17, 261, doi: [10.1038/s41592-020-0772-5](https://doi.org/10.1038/s41592-020-0772-5)